

Agent-based modelling of green gentrification around parks and other urban green spaces taking into account cost implications

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Abstract

Urban green spaces provide substantial ecological, social, and economic benefits, yet growing evidence shows that greening interventions can trigger green gentrification, leading to rising property values and the displacement of lower-income residents. This work examines the feedback mechanisms and economic thresholds of green gentrification induced by urban green space interventions, with particular focus on establishment and maintenance costs. Existing research predominantly identifies whether gentrification occurred rather than explaining the underlying feedback mechanisms and thresholds, and the role of park costs has been largely overlooked. This work addresses these gaps by developing an agent-based model that extends an existing dynamic gentrification framework through the explicit integration of urban green spaces and their associated costs. The model simulates heterogeneous household agents that relocate across a grid-based urban environment based on two main drivers: rents that adjust endogenously to demand, and location attractiveness, which combines park proximity, quality, size, and function into a single composite score. Establishment and maintenance costs are incorporated explicitly, capturing how the economic burden of green infrastructure feeds back into rent dynamics and, ultimately, into residential displacement. A structured experimental evaluation shows that the timing of tipping points follows a monotonic, though not sharply thresholded, relationship with park proximity, quality, size, and function, with proximity exerting the strongest and most linear effect and quality and size showing diminishing marginal returns. The probability of reaching a tipping point, by contrast, cannot be attributed to any single attribute threshold, but instead emerges from interaction effects between park characteristics. A comparable asymmetry appears in the economic dimension: establishment costs, tied to park size, primarily determine how quickly a tipping point is reached, while maintenance costs, through their role in sustaining park quality over time, more strongly influence its likelihood. These findings extend prior agent-based gentrification research by showing that green gentrification is best understood as an emergent process shaped jointly by the ecological and economic characteristics of green infrastructure, rather than by a single universal threshold. The distinction between a comparatively predictable timing and an interaction-dependent probability of tipping offers a first step toward operationalizing the "just green enough" planning paradigm, suggesting that mitigation requires attention to the specific combination of park attributes and cost structure rather than any isolated characteristic. The model is best understood as an exploratory decision-support framework, and its findings should be interpreted relative to the simplifications inherent in the simulated environment, cost structure, and operational tipping-point definition.

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List of Abbreviations

ABM	Agent-Based Model(ling)
CBA	Cost-Benefit Analysis
CI	Confidence Interval
GIS	Geographic Information System
GSA	Global Sensitivity Analysis
H1	Hypothesis 1
H2	Hypothesis 2
HP	Hedonic Pricing
IQR	Inter-quartile Range
JGE	Just Green Enough
MCDA	Multi-Criteria Decision Analysis
RQ1	Research Question 1
RQ2	Research Question 2
SES	Social-Ecological System(s)
UGS	Urban Green Space(s)

1. Introduction

Urbanization has increasingly reshaped the relationship between cities, nature, and society, making urban green infrastructure a key element of sustainable urban development. Parks, green corridors, and other forms of urban greenery are widely regarded as essential components of livable, climate-resilient cities, and municipalities worldwide continue to invest substantially in expanding and upgrading their green infrastructure. While green spaces provide important ecological, social, and economic benefits, they can also trigger unintended socio-spatial consequences such as green gentrification, whereby the very interventions designed to improve neighborhoods contribute to rising housing costs and the displacement of long-term, often lower-income, residents. This apparent contradiction between environmental improvement and social equity lies at the heart of the present work and motivates the development of a simulation-based approach capable of capturing how such dynamics unfold over time. This chapter introduces the motivation and research context of this work by discussing the challenges associated with balancing environmental improvements, economic impacts, and social equity. It further identifies existing research gaps, defines the objectives, research questions, and hypotheses of this study, and provides an overview of the thesis structure and the development of an agent-based modeling framework to analyze green gentrification dynamics.

1.1 Motivation

Urban green spaces are increasingly promoted as cost-effective nature-based solutions that enhance environmental quality, public health, and urban resilience [32]. However, growing empirical evidence shows that investments in urban greening can trigger green gentrification, leading to rising property values, displacement pressures, and uneven distribution of both social benefits and cost burdens [3]. This tension highlights a critical knowledge gap regarding how green infrastructure investments, associated costs, and socio-spatial dynamics interact over time. Addressing this gap is becoming increasingly important as cities worldwide invest heavily in urban green infrastructure to simultaneously pursue climate adaptation, biodiversity conservation, and improved quality of life. While such interventions provide substantial environmental, social, and economic benefits, they also raise fundamental questions about social equity and the fair distribution of these benefits among different population groups [36]. Existing research has predominantly relied on empirical or statistical analyses that identify green gentrification after it has occurred, offering limited insight into the dynamic feedback mechanisms and critical thresholds that drive these processes [10, 36]. Agent-based modeling provides a promising approach to overcome these limitations, as it enables the simulation of heterogeneous household decisions, housing market dynamics, and neighborhood transformation

as interacting processes from which green gentrification emerges endogenously over time [7, 30]. Furthermore, explicitly incorporating establishment and maintenance costs into such models extends current research beyond the traditional focus on environmental amenities alone, allowing urban planning strategies to be evaluated not only in terms of ecological effectiveness but also regarding their long-term economic feasibility and social consequences [34].

1.2 Problem Statement

While the existing literature has established a strong empirical relationship between urban green spaces and green gentrification, it predominantly focuses on identifying whether green gentrification occurs rather than explaining how it emerges over time. Most studies rely on retrospective statistical analyses, hedonic pricing models, or on geographic information systems based approaches that capture observed outcomes but provide only limited insight into the underlying feedback mechanisms, residential decision-making, and non-linear neighborhood transformation processes that drive green gentrification [9, 10, 36]. Consequently, the critical thresholds at which environmental improvements evolve into self-reinforcing gentrification dynamics remain poorly understood [33].

One prominent attempt to address these challenges is the Just Green Enough (JGE) approach proposed by Curran and Hamilton, which advocates targeted, small-scale, community-oriented greening interventions to achieve ecological and public health benefits while minimizing displacement pressures and excessive increases in property values [16]. Although the concept has become an influential planning paradigm, it remains largely normative. Its central principle of implementing interventions that are "just green enough" lacks operational criteria, making it difficult to quantify, evaluate empirically, or translate into practical planning guidelines across different urban contexts [16, 36]. A further limitation of the existing literature is the limited consideration of the economic dimension of urban green infrastructure. While numerous studies examine how environmental amenities influence housing markets, comparatively little attention has been paid to how the establishment and long-term maintenance costs of urban green spaces shape neighborhood attractiveness and potentially contribute to green gentrification. Yet these costs determine not only the quality and persistence of green infrastructure but also the financial resources required to sustain it, thereby influencing both environmental outcomes and housing affordability [14, 34]. Moreover, current research rarely integrates these cost implications into dynamic modeling approaches. Existing studies predominantly rely on empirical observations or static valuation methods, providing little support for evaluating alternative investment strategies before implementation [10, 34]. Consequently, urban planners still lack simulation-based decision support tools that jointly consider environmental quality, socio-spatial dynamics, and the economic implications of urban green infrastructure.

1.3 Goal Setting

To address the research gaps identified above, this work develops an agent-based modeling framework for analyzing the emergence of green gentrification around urban green spaces. In contrast to predominantly empirical and retrospective approaches, the proposed framework aims to capture the dynamic interactions between heterogeneous households, urban green spaces, housing market mechanisms,

and neighborhood transformation processes. By explicitly representing residential decision-making and endogenous feedback loops, the model seeks to improve the understanding of how incremental environmental improvements can develop into self-reinforcing green gentrification dynamics.

A particular focus of this work is the integration of the economic dimension of urban green infrastructure. Besides modeling the environmental attractiveness of parks, the framework explicitly incorporates both establishment and long-term maintenance costs, allowing the socio-spatial consequences of different investment strategies to be assessed. In doing so, the model extends existing approaches by combining environmental, economic, and social processes within a unified simulation framework. This enables the investigation of planning scenarios that cannot easily be studied using conventional empirical methods and provides a basis for identifying critical thresholds at which green infrastructure investments begin to produce unintended social consequences. To this end, the model is evaluated through a series of simulation experiments that systematically vary park characteristics and cost scenarios across many stochastic repetitions, allowing tipping dynamics and their sensitivity to different parameters to be assessed in a statistically robust manner.

To achieve these objectives, two research questions are addressed. Firstly,

RQ1: At what thresholds of park quality, size, or proximity does ecological enhancement tip over into a self-reinforcing gentrification dynamic?

Secondly,

RQ2: How do establishment and maintenance costs for urban parks influence the tipping points at which green upgrades have a gentrifying effect?

For answering the second research question, two different hypotheses are formulated. Hypothesis 1 (H1) assumes that higher establishment costs, which are typically associated with larger or higher-quality green infrastructure projects, lower the threshold of environmental attractiveness at which green gentrification emerges. Hypothesis 2 (H2) assumes that higher maintenance costs contribute to preserving park quality over time and therefore shift green gentrification from a short-term phenomenon towards a more stable long-term process. These hypotheses, together with RQ1, are examined through the corresponding simulation experiments. Ultimately, the developed framework is intended as a simulation-based decision-support tool for sustainable urban planning. By enabling the ex-ante evaluation of alternative park designs and investment strategies, it supports planners in balancing ecological benefits, economic efficiency, and social equity while reducing the risk of unintended displacement and green gentrification.

1.4 Outline

The remainder of this work is organized as follows. Chapter 2 establishes the theoretical foundation of the study by introducing the concepts of urban green spaces, green gentrification, agent-based modeling, tipping points, and the cost implications of urban green infrastructure. Furthermore, it reviews the existing literature on green gentrification and discusses two recent agent-based modeling approaches that provide the conceptual basis for the framework developed in this work. Building upon these theoretical and methodological foundations, Chapter 3 presents the

proposed agent-based model in detail. It describes the overall model design, the behavioral mechanisms of the agents, the representation of urban green spaces, and the parameterization based on empirical findings from the literature. Subsequently, Chapter 4 translates the conceptual model into its technical implementation, explaining the architecture of the simulation framework as well as the realization of the individual model components and their interactions. Based on the implemented model, Chapter 5 evaluates the proposed framework through a series of simulation experiments. It first introduces the experimental design developed to address the research questions, followed by the presentation and analysis of the simulation results, including a global sensitivity analysis, the identification of green gentrification tipping points through a full-factorial parameter sweep, and the investigation of different cost scenarios. Finally, Chapter 6 synthesizes the findings by summarizing the work, answering the two research questions in detail, and evaluating the proposed hypotheses in light of the results, thereby relating the findings back to the problem statement and goal setting formulated in Chapter 1. Building on this synthesis, the chapter derives policy recommendations for sustainable urban planning, before discussing the limitations of the developed framework and outlining corresponding directions for future research.

2. Theoretical Background and Related Work

Building on the motivation, problem statement, and research questions outlined above, this chapter establishes the conceptual and methodological groundwork required to develop the agent-based green gentrification model presented in Chapter 3. Before the model itself can be designed, it is necessary to clarify the theoretical concepts it builds upon, to understand the mechanisms through which environmental improvements can translate into socio-spatial displacement, and to position the present work within the existing body of agent-based gentrification research. This chapter therefore provides the theoretical foundation for the developed green gentrification model and reviews the most relevant related work in this research field. First, the concepts of urban green spaces and green gentrification are introduced, including their environmental, social, and economic implications for urban transformation processes; this establishes the conceptual vocabulary and empirical background against which the model's assumptions and parameterization are later justified. Subsequently, the chapter discusses agent-based modeling and tipping point theory as methodological foundations for simulating complex socio-spatial dynamics, thereby motivating the choice of agent-based modeling as the methodological approach pursued in this work and introducing the conceptual understanding of tipping points that underlies the operational tipping-point definition used in the experimental evaluation. Finally, two existing agent-based approaches of gentrification models are examined, namely by Mauro et al. [31] and Caprioli et al. [10], which form the conceptual basis of the present study.

2.1 Urban Green Spaces

According to Haase et al. [21] and Vidal et al. [43] UGS are defined as urban land, partly or completely covered with grass, trees, shrubs, or vegetation. They provide critical ecosystem services, such as parks, forests, green roofs, streams, and community gardens. Numerous empirical studies have shown that UGS positively affect the quality of life for residents and are integrated into the everyday lives and life courses of urban populations [35]. They have not only an important role in responding environmental challenges by absorbing CO₂ and releasing O₃, decreasing temperatures in cities and improving air quality [37, 45]. But also from a social perspective by enhancing human physical health and mental well-being by providing a safe place for sport activities and offering an appropriate environment for psychological restoration and feeling well [25]. Furthermore, the benefits of UGS also include economic advantages as they generate job opportunities, attract business and tourism and minimize health expenses [32]. In general the overall quality of life perceived by

citizens is positively correlated with urban greenness, and even small increases in greenness can improve the quality of life especially for those citizens with lower per capita income [19]. Quality takes precedence over quantity here: high-quality, easily accessible, and well-maintained green spaces can significantly increase the well-being of city dwellers [47].

However, some studies show that UGS can also negatively affect human well-being. In general, urban green space of bad quality can have negative effects. For example, the lack of richness of biodiversity in UGS and also UGS located next to roads (urban greenways), which lead to environmental stressors such as traffic noise or air pollution, can negatively affect human physical and psychological well-being, especially for older people [12]. Another negative effect is the displacement of users between urban green spaces, which can occur when use levels are high, although in larger sites visitors often redistribute within the same area. Evidence suggests a social carrying capacity of about 1 person per hectare per hour (approx. 10 per hectare per day), while very low levels of use are associated with increased feelings of fear and concerns about personal safety [1]. And finally, if residents fear crime in their neighborhood, the positive life satisfaction effects of access to greenspace are almost completely forgone [2, 18].

2.2 Green Gentrification

Despite the widely documented benefits, the socio-spatial consequences of greening interventions remain highly uneven. Green gentrification is the resulting effect, when improving an area's attractiveness by repairing damaged urban settings, developing greenspace, or deploying climate-adaptive green infrastructure, lead to rising property values, housing costs, and the physical displacement of working-class inhabitants and racialized groups and cultures [3]. It creates enclaves of environmental privilege, where groups of residents are pushed out of their neighborhoods. These groups are often low-income households, ethnic minorities, migrants or elderly residents with limited mobility or fixed incomes [4]. Green gentrification is hereby not an accidental side effect but is closely linked to neoliberal urban governance, city branding and competitiveness strategies or market-oriented planning approaches [3]. Generally, it produces measurable economic gains, where property owners capture most benefits, so the aggregated gains are not the same as the universal welfare gains [28]. The increased property values around parks are estimated to be between 1.8 and 3.9-times the ongoing costs of the green space [5].

While the existing literature primarily focuses on investigating whether gentrification occurs as a result of urban green spaces, comparatively little attention has been paid to the underlying dynamics and possible mitigation strategies of this development. One of the most prominent counter-concepts in this context is the JGE approach proposed by Curran and Hamilton [16]. The concept emerged as a critique of large-scale flagship greening projects that often prioritize urban branding, real estate valorization, and competitiveness over social equity. According to JGE, urban greening should not aim to maximize aesthetic attractiveness or economic value enhancement, but rather provide environmental improvements that directly benefit existing local communities without triggering speculative investment and displacement pressures. Instead of implementing highly visible and capital-intensive flagship parks, waterfront redevelopments, or iconic green infrastructure projects,

JGE promotes small-scale, community-oriented, and functionally modest green interventions. These include, for example, community gardens, localized pocket parks, ecological restoration projects, or environmentally improved industrial landscapes that primarily serve the needs of current residents. The central idea is to achieve ecological and public health benefits while avoiding the excessive increase in land and housing values that often accompanies large prestige green projects. From a socio-spatial perspective, the approach can be understood as a potential strategy to reduce displacement risks and preserve social diversity within urban neighborhoods. By limiting large-scale symbolic value enhancement and focusing on locally embedded environmental improvements, JGE attempts to slow down the rent escalation and socio-economic restructuring processes commonly associated with green gentrification. Furthermore, the concept emphasizes participatory planning processes and community involvement, thereby strengthening local ownership and ensuring that environmental investments primarily benefit long-term residents rather than external investors or higher-income newcomers. However, despite its strong normative appeal, the practical implementation of the JGE approach remains challenging. Critics argue that even smaller-scale greening interventions may eventually contribute to rising neighborhood attractiveness and housing demand, particularly in highly competitive urban housing markets. Nevertheless, the concept provides an important theoretical and policy-oriented framework for discussing how environmental sustainability goals and social justice objectives can be balanced within urban development processes.

2.3 Agent-based Models

Understanding urban transformation dynamics such as green gentrification requires analytical approaches capable of capturing heterogeneous actors, local interactions, feedback mechanisms, and long-term spatial change. ABM provides a powerful methodological framework for simulating these complex socio-spatial processes through a bottom-up perspective [8, 20]. Rather than describing system behavior through aggregate equations, ABM models complex systems as collections of autonomous decision-making entities (agents) whose individual actions and interactions give rise to system-level outcomes. This bottom-up paradigm is based on the principle of **emergence**, whereby macro-scale patterns arise from numerous micro-level decisions without being explicitly programmed into the model [20]. Consequently, ABM is particularly well suited for studying complex adaptive systems in which global dynamics cannot be understood solely from aggregate relationships.

An agent is typically defined as an autonomous computational entity that perceives its environment, makes decisions according to predefined behavioral rules, and performs actions to achieve specific objectives [8]. Depending on the research question, agents may represent households, residents, investors, developers, policymakers, or other stakeholders. Besides possessing individual attributes such as income, preferences, or location, agents are characterized by several fundamental properties. They are **autonomous**, meaning that they make decisions independently without centralized control; **reactive**, as they respond to changes in their surrounding environment; **proactive**, because they pursue individual goals based on internal decision rules; and **social**, as they interact with other agents through communication, competition, cooperation, or the exchange of information and resources [31]. Through these interactions, agents continuously adapt their behavior to changing environmental and

social conditions, allowing the model to capture dynamic feedback processes and path-dependent development trajectories.

The environment constitutes the second fundamental component of an ABM. It provides the spatial, social, institutional, and ecological context within which agents operate and interact. Depending on the application, the environment may consist of geographical space, housing markets, transportation infrastructure, urban green spaces, policy regulations, or other contextual variables that influence agent behavior. Importantly, the relationship between agents and their environment is bidirectional: while agents respond to environmental conditions, their collective actions simultaneously modify the environment, creating feedback loops that drive system evolution over time (see Fig. 2.1) [20]. This reciprocal interaction between agents and their environment distinguishes ABM from many traditional modeling approaches and enables the representation of complex adaptive systems characterized by non-linearity, feedback mechanisms, and self-organization.



Figure 2.1: Relationship of agent and environment in ABM.

Due to its flexibility and comparatively simple conceptual structure, ABM has been widely applied across numerous scientific disciplines, including physics, economics, geography, ecology, and the social sciences [11, 13, 15]. Within urban studies, agent-based models are increasingly employed to investigate urbanization processes, residential mobility, segregation dynamics, land-use change, housing markets, and broader socio-economic transformation processes. Their ability to explicitly represent heterogeneous decision-makers, spatial interactions, and adaptive behavior makes ABM particularly suitable for analyzing how local household decisions, neighborhood effects, and policy interventions jointly generate emergent urban patterns over time. Furthermore, Vallejo et al. [42] demonstrate how ABM can be applied to optimize the spatial distribution of urban green spaces by representing diverse stakeholders and complex land-use trade-offs. According to the authors, ABM provides a valuable decision-support tool for evaluating sustainable urban planning strategies and improving the understanding of interactions between environmental amenities, land-use decisions, and socio-economic urban dynamics.

2.4 Tipping Points

In Social-Ecological System(s) (SES), complex interactions between human activities and ecological processes create feedback loops that can drive large-scale system transformations [27]. These systems may approach critical thresholds, commonly referred to as **tipping points**, beyond which relatively small disturbances can trigger disproportionate, non-linear changes that shift the system from one stable state to another (e.g., from a "healthy" to a "degraded" ecosystem). According to Milkoreit

et al. (2018) [33], a tipping point should not be understood merely as a threshold itself, but rather as the process by which crossing a critical threshold initiates a self-reinforcing transition towards an alternative system state. This process is typically driven by positive feedback mechanisms, causing system responses that are no longer proportional to the initial disturbance.

A key characteristic of tipping points is their non-linearity: minor incremental changes in system drivers can suddenly lead to large and often unexpected transformations. Although tipping points are frequently associated with irreversible change, hereby irreversibility is not a defining characteristic [33]. Instead, systems may remain theoretically reversible while being practically difficult or prohibitively costly to restore because of hysteresis. Consequently, simply removing or reversing the original pressures is often insufficient to return the system to its previous state. Unlike purely ecological systems, tipping dynamics in SES are further complicated by human decision-making, institutional arrangements, governance structures, economic incentives, and changing social norms. These social processes interact with ecological dynamics through multiple feedback loops, making the timing and location of tipping points inherently difficult to predict. Although various early warning indicators have been proposed, their interpretation remains challenging due to the complexity and interconnectedness of social and ecological processes.

Importantly, tipping points are not inherently negative. While undesirable tipping points may result in ecosystem degradation, biodiversity loss, or social displacement, desirable tipping points can accelerate sustainability transitions, such as the rapid adoption of renewable energy or ecosystem restoration [33]. This broader understanding is particularly relevant in the context of urban green infrastructure and green gentrification, where environmental improvements may generate self-reinforcing ecological benefits while simultaneously initiating positive socioeconomic feedback, including rising property values, increased investment, and residential displacement. Understanding these dynamics requires interdisciplinary approaches that integrate ecological, social, and economic processes to identify critical thresholds and support more sustainable urban governance.

2.5 Cost Implications

From an economic perspective, the costs associated with urban green infrastructure are commonly distinguished into **establishment costs** and **maintenance costs**. Establishment costs refer to the one-time expenditures required for planning, site preparation, construction, planting, and installation, whereas maintenance costs comprise the recurring expenses necessary to ensure the long-term functionality, ecological performance, and aesthetic quality of green infrastructure, including activities such as irrigation, pruning, mowing, replacement planting, and general upkeep [46]. Considering both cost categories is essential for assessing the financial implications of UGS, as maintenance requirements can significantly influence the overall life-cycle costs of green infrastructure.

Existing research consistently demonstrates that both establishment and maintenance costs vary considerably across different types of urban green infrastructure. Cost differences are primarily driven by the intervention typology, vegetation characteristics, spatial scale, system complexity, and local geographic conditions. Moreover, maintenance costs generally increase with establishment costs, indicating that

more complex and multifunctional greening interventions require greater long-term financial commitments [6]. Providing empirical reference values, Panduro et al. estimate mean establishment costs for urban green spaces at 59.98€/m² and mean annual maintenance costs at 1.79€/m², while also emphasizing the substantial variability across projects [34]. Despite these costs, the literature increasingly frames urban green infrastructure as a long-term investment rather than a financial burden. Establishment and maintenance expenditures contribute to enhanced urban quality of life [14] and are often offset by a wide range of ecosystem services and socioeconomic benefits. These include urban cooling, oxygen production, heat stress reduction, energy savings, and other regulating ecosystem services, whose aggregated economic value may substantially exceed implementation and maintenance costs [6]. Depending on the combination of greening measures, estimated annual economic benefits range from approximately £10 to more than £800 per square meter, with complex, multi-layered green infrastructure systems generating the highest returns [6].

However, assessing the economic performance of NBS remains methodologically challenging because many of their environmental and social benefits are difficult to express in monetary terms. Conventional Cost-Benefit Analysis (CBA) therefore face limitations, as they require the aggregation of heterogeneous ecosystem services into a single monetary metric and often fail to capture the full range of co-benefits associated with nature-based interventions. This complicates comparisons between alternative NBS as well as between green and conventional grey infrastructure. To overcome these shortcomings, Panduro et al. propose a stepwise cost-effectiveness approach that prioritizes interventions according to policy objectives rather than relying exclusively on monetary valuation. This approach provides a more transparent and practically applicable framework for evaluating and prioritizing urban green infrastructure investments [34].

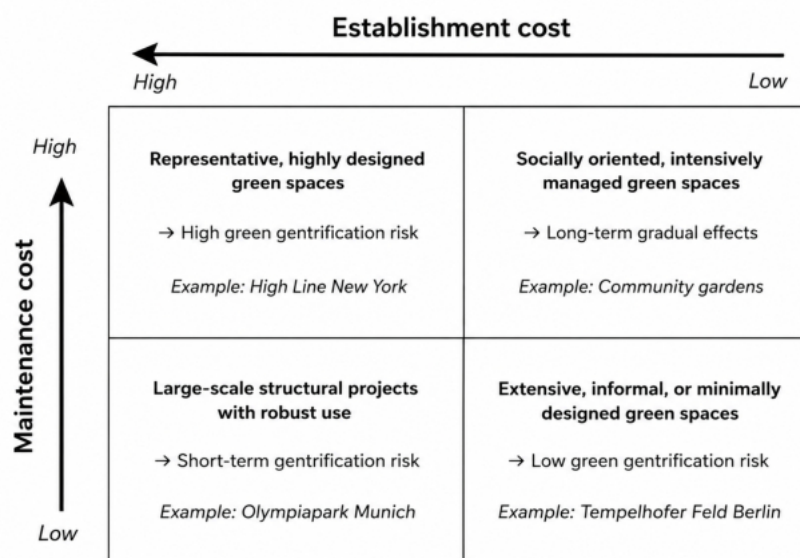


Figure 2.2: Cost scenarios of UGS based on establishment and maintenance costs (AI-generated graphic).



(a) New York High Line

Source: gocity.com (2025)



(b) Olympiapark München

Source: hotel-rothof.de (2026)



(c) Community Garden Berlin

Source: berlin.de (2026)



(d) Tempelhofer Feld Berlin

Source: berlin.de (2026)

Figure 2.3: Examples of UGS representing different cost scenarios.

The financial implications of urban green infrastructure can be further differentiated by considering the relationship between establishment and maintenance costs. Different combinations of these cost categories are typically associated with distinct design approaches, management intensities, and socio-spatial outcomes. From the perspective of green gentrification, these cost profiles may also provide insights into the likelihood and timing of neighborhood change. A short overview of the four different scenarios is given in Fig.2.2 and some examples of them can be seen in Fig. 2.3. A first scenario comprises interventions characterized by **high establishment and high maintenance costs**. These projects typically involve highly designed, iconic public spaces with intensive landscaping, high-quality materials, and continuous management to preserve their visual appearance and functionality. Such investments often create significant place-based amenities that attract higher-income residents, private investment, and tourism, thereby increasing the risk of green gentrification. The *High Line in New York City* (Fig. 2.3a) represents one of the most prominent examples of this development trajectory, where a flagship green infrastructure project became closely associated with rising property values, commercial redevelopment, and residential displacement [29]. Similar cost structures can also be observed in highly maintained recreational landscapes such as urban golf courses.

A second scenario consists of projects with **high establishment costs but relatively low maintenance costs**. These interventions require substantial upfront investments for construction and landscape transformation but are subsequently characterized

by comparatively low operational expenditures due to robust design, durable infrastructure, or extensive vegetation management. Large urban parks developed as part of strategic regeneration projects often fall into this category. While these interventions may trigger considerable increases in land and property values shortly after implementation, their long-term maintenance burden remains comparatively moderate. The *Olympiapark in Munich* (Fig. 2.3b) illustrates such a cost profile, where major initial investments created a multifunctional green space that requires relatively limited structural adaptation over time.

The third scenario includes interventions with **low establishment costs but high maintenance costs**. These projects are often implemented with limited initial investment but rely on continuous management, programming, and maintenance to achieve their intended social and ecological functions. Neighborhood parks, community gardens, and other socially oriented green spaces frequently require regular horticultural care, programming, and community engagement despite relatively modest construction costs (Fig. 2.3c). Their influence on neighborhood attractiveness may therefore emerge gradually through sustained improvements in environmental quality and social cohesion, potentially contributing to slower and more incremental forms of green gentrification.

Finally, interventions characterized by **low establishment and low maintenance costs** generally comprise extensive, informal, or minimally designed green spaces that rely largely on natural vegetation dynamics and require only limited management. Examples include temporary green spaces on vacant land, informal urban meadows, or large open landscapes such as the *Tempelhofer Feld in Berlin* (Fig. 2.3d). Such interventions align closely with the concept of JGE, which advocates for environmental improvements that enhance ecological quality and public accessibility without creating highly marketable landscapes that stimulate speculative real estate investment. As a result, these projects are generally considered to pose a comparatively low risk of green gentrification while still delivering important ecological and recreational benefits.

2.6 Related Work

Recent research on green gentrification increasingly emphasizes the importance of dynamic and spatially explicit modeling approaches to better understand the interaction between urban green spaces, residential mobility, and socio-economic transformation processes [4, 36]. While early agent-based gentrification models primarily focused on income segregation, housing markets, and neighborhood change [17, 26, 40], more recent studies integrate environmental amenities and green infrastructure as central drivers of urban transformation [9, 10]. At the same time, most existing green gentrification research still relies predominantly on empirical Geographic Information System (GIS) analyses, hedonic pricing models, or statistical approaches [36, 45], which often lack the ability to simulate long-term feedback dynamics and emergent socio-spatial processes. Consequently, there remains a significant research gap regarding dynamic agent-based models that explicitly combine residential decision-making, endogenous rent formation, and environmental attractiveness within a unified simulation framework. As the underlying concept of the proposed green gentrification model is based on the two recently published approaches

of Mauro [31] and Caprioli [10], the following two sub-sections taking a deeper look into those papers.

2.6.1 Mauro et al. 2025

Firstly, Mauro et al. [9] present an agent-based model that illustrates the emergence and evolution of gentrification in cities. The urban environment is modeled as a grid layout, with each cell representing a district. This grid is populated with different agents (households), which are divided into three socioeconomic groups: low income (L), middle income (M), and high income (H). These households relocate between neighborhoods based on living costs and economic conditions and a simulation shows how their relocation decisions reshape urban neighborhoods over time. L accounts for 38% of the population, M for 57%, and H for the remaining 5%. Low-income agents move when priced out of a neighborhood; medium-income agents gravitate towards areas with similar economic conditions and quality of life; and high-income agents are attracted to areas undergoing economic growth, where they can maximize their investment returns. Gentrification only emerges when high-income residents have some mobility. Fig. 2.4 shows the grid layout and income and spatial distribution.

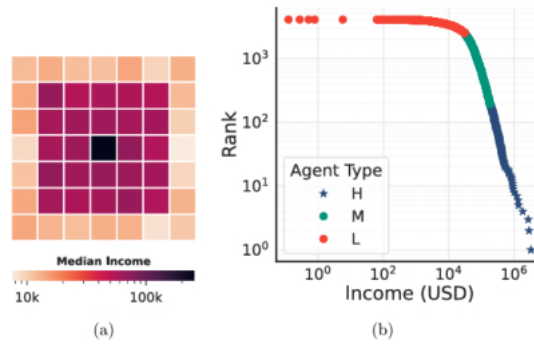


Figure 2.4: Income distribution in grid-layout (a) and spatial distribution (b) [31].

In order to analyze the dynamics of neighborhood change, the authors introduce a temporal network-based indicator that tracks the influx of wealthier residents and the outflow of lower-income residents. This approach enables the model to detect gentrification processes at an earlier stage than traditional indicators based on population counts or demographic change. Simulations demonstrate how economic interactions between different income groups generate spatial patterns resembling real-world gentrification processes. The model also shows that urban density significantly accelerates gentrification dynamics by increasing interactions and competition for housing in denser urban structures.

The model provides a useful baseline framework for studying gentrification dynamics. Its high level of abstraction enables the core mechanisms of income-based relocation and housing price feedback to be analyzed clearly. Another advantage is that the full model code is openly accessible, improving transparency and reproducibility while enabling further extensions by other researchers.¹ However, the model does not

¹The full code of Mauro et al. can be found at: <https://github.com/maurucz/Gentrification>

incorporate green spaces or other urban amenities, nor is it calibrated with real-world data, which limits its ability to represent specific urban contexts or produce empirically validated predictions.

2.6.2 Caprioli et al. 2023

Secondly, in their paper, Caprioli et al. [10] propose a hybrid modeling framework that combines agent-based modeling, Multi-Criteria Decision Analysis (MCDA) and Hedonic Pricing (HP) to examine the dynamics of green gentrification around urban green spaces. The study focuses particularly on the socio-economic impacts of transforming a former golf course into a large urban park in Brisbane, Australia. The central objective of the model is to investigate how environmental improvements influence housing prices, residential mobility, and neighborhood socio-spatial composition over time. An overview of this mixed-methods approach can be seen in Fig. 2.5.

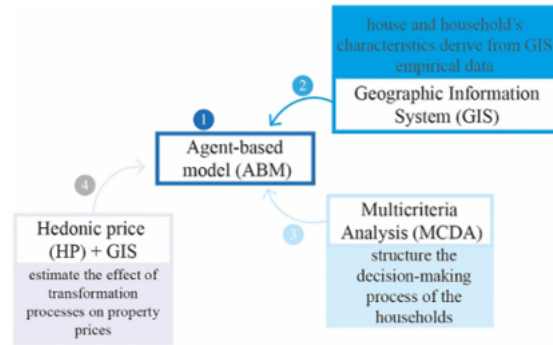


Figure 2.5: Overview of the approach of Caprioli et al. [10].

Within the framework, the ABM component is used to simulate residential mobility and neighborhood transformation processes through individual household decision-making. Households are represented as heterogeneous agents with different socio-economic characteristics and residential preferences. Agents continuously evaluate alternative residential locations based on housing affordability, accessibility, environmental attractiveness, and neighborhood quality. Residential relocation therefore emerges endogenously from decentralized interactions between households and their surrounding urban environment. This allows the model to reproduce dynamic socio-spatial processes such as displacement, segregation, and neighborhood upgrading. The MCDA component is used to evaluate the environmental attractiveness of urban green spaces and surrounding neighborhoods. Caprioli et al. integrate multiple spatial, environmental, and socio-economic criteria into a composite attractiveness index. These criteria include factors such as park accessibility, ecological quality, proximity to services, transport infrastructure, and neighborhood amenities. The weighting of these criteria is determined through stakeholder and expert participation, enabling the incorporation of local planning priorities and contextual urban knowledge into the simulation. MCDA therefore serves as a decision-support framework that translates qualitative urban planning assessments into quantifiable attractiveness values used within the ABM. In addition, the model incorporates a spatial HP approach to estimate the influence of environmental amenities on housing prices. Hedonic pricing assumes that real estate values reflect the capitalization

of neighborhood characteristics and environmental qualities. Within the Caprioli framework, spatial HP is used to quantify how proximity to parks and green infrastructure affects local property values. This component creates a direct connection between environmental attractiveness and housing market dynamics by translating park-related amenities into economic value increases.

The simulation results demonstrate that large-scale green infrastructure projects can significantly increase neighborhood attractiveness and property values, particularly in areas located close to the newly developed urban park. As a consequence, higher-income households become increasingly attracted to these neighborhoods, while lower-income residents experience growing affordability pressure and displacement risks. The model therefore illustrates how environmental improvements may unintentionally contribute to socio-spatial inequality and green gentrification processes. Furthermore, the results highlight that the magnitude of gentrification effects strongly depends on the spatial configuration of green spaces, local housing market conditions, and stakeholder-defined planning priorities.

The approach offers several important advantages. Its comparatively low level of abstraction and explicit integration of Urban Green Spaces allow the model to analyze concrete planning scenarios and produce policy-relevant insights for sustainable urban development. In particular, the combination of ABM, MCDA, and HP enables the simultaneous representation of social behavior, environmental valuation, and housing market dynamics within a unified simulation framework. However, the detailed representation of urban processes also increases overall model complexity, which may reduce transparency, transferability, and generalizability. In addition, the model code is not publicly available, limiting reproducibility and restricting the ability of other researchers to adapt or further extend the framework.

3. Methodology

Having established the theoretical foundations of urban green spaces, green gentrification, agent-based modeling, and tipping point theory, as well as the two related agent-based approaches of Mauro [31] and Caprioli [10], this chapter now translates these conceptual and methodological building blocks into the concrete model developed for this work. Methodologically, the study follows a design-science approach centered on the iterative construction, parameterization, and refinement of a simulation artifact, rather than the statistical analysis of an existing empirical dataset. Instead of observing gentrification after it has occurred, this approach makes it possible to generate socio-spatial dynamics endogenously from a set of explicitly formulated behavioral rules and to systematically explore how changes in individual model components affect the resulting system-level outcomes. Concretely, an agent-based simulation model is constructed that extends Mauro’s dynamic gentrification framework through the explicit integration of urban green spaces, inspired by Caprioli’s MCDA-based environmental valuation, and further incorporates park establishment and maintenance costs as an additional economic dimension that has been largely absent from prior agent-based gentrification research. Model design therefore proceeds by conceptually combining and extending established modeling components rather than developing every mechanism from first principles, allowing the present work to build on already validated modeling logic while introducing the environmental and economic extensions. Once the conceptual model has been specified, its parameters are grounded empirically wherever possible, drawing on official statistical data (e.g., income distributions, median rents) as well as empirical cost estimates and weighting schemes reported in the green gentrification and urban green infrastructure literature. The following chapter presents the methodological framework of this study and provides a comprehensive description of the developed agent-based green gentrification model. First, the conceptual model design and its underlying theoretical mechanisms are explained. Subsequently, the parameterization and calibration process is outlined, including the integration of literature-based assumptions and empirical real-world data.

3.1 Model Design

Taking the fundamentals from the previous chapter into account, this study develops an agent-based model of urban green gentrification, simulating the interaction between socio-economic residential dynamics and environmental amenities in urban spaces. A complete overview of the research concept draft can be seen in Fig. 3.1. The model investigates how UGS influence neighborhood attractiveness, residential mobility, rent escalation, and socio-spatial transformation over time. The overall framework combines concepts from urban economics, environmental valuation, and

dynamic gentrification modeling. As mentioned in the chapter above, the model builds conceptually upon the dynamic gentrification approach proposed by Mauro [31] and extends it through the integration of environmental amenity valuation using MCDA, inspired by Caprioli [10] as well as the green gentrification literature of Rigolon and Bottero [9, 36].

The central objective of the model is to reproduce green gentrification as an emergent process resulting from the interaction between heterogeneous household agents, dynamic housing costs, residential relocation, neighborhood composition and environmental attractiveness generated by urban green infrastructure. Therefore, the city is represented as a spatially explicit **grid-based urban environment**, as proposed by Mauro, in which each grid cell represents a simplified neighborhood unit characterized by residential population, housing costs, occupancy level and environmental attractiveness. The model posits the hypothesis that the dynamics of urban environments are derived from interactions between individual households and their immediate local neighborhood environments, which are characterized by decentralization. Moreover, the population of the city grid consists of heterogeneous household agents divided into three income groups (low-, middle- and high-income households), with each group exhibiting a distinct income distribution. These agents are tasked with the continuous evaluation of residential conditions, and the subsequent determination of relocation decisions based on a range of socio-economic and environmental preferences. The present model additionally assigns continuous income values to agents based on empirical income distributions. This enables a more differentiated representation of purchasing power and rent sensitivity. The socio-economic composition of neighborhoods is subject to dynamic change over time as households relocate between urban areas. Consequently, the transformation of neighborhoods is driven by endogenous migration processes rather than by predefined redevelopment interventions. This approach is in alignment with the prevailing methodological logic espoused by Mauro, wherein the process of gentrification is not externally imposed but rather emerges endogenously through dynamic feedback processes between households and neighborhoods. It adapts the underlying grid layout and the three distinct income groups.

The **housing prices** in this environment evolve dynamically during the simulation and are determined endogenously by local demand and environmental attractiveness. Hereby, rents increase due to two principal mechanisms: increasing residential demand and capitalization of environmental amenities. Consequently, neighborhoods with higher occupancy levels and greater environmental attractiveness become more expensive over time. This operationalizes the core mechanism of gentrification, namely the gradual exclusion of lower-income households through rising housing costs. Compared to Mauro, the present model extends rent formation through explicit hedonic pricing of UGS, inspired by the approach of Caprioli (2023) [10]. Environmental amenities therefore become a direct driver of housing market transformation. These endogenous rent dynamics directly influence **household decision-making**, which is modeled as a utility-maximizing process and therefore constitutes the economic foundation of residential mobility. Households evaluate their current neighborhood conditions and compare them with alternative residential opportunities. Utility depends on several interacting dimensions, including remaining disposable income after rent payment, the burden of rent relative to income, environmental attractiveness, and social neighborhood composition. Households pre-

fer affordable neighborhoods with attractive green amenities and socially similar neighbors. If the utility of the current residential location falls below a predefined threshold, households search for alternative locations and relocate to neighborhoods with higher expected utility. The utility function further incorporates differentiated group sensitivities to housing costs, whereby low-income households exhibit substantially higher vulnerability to rising rents than high-income households. This mechanism operationalizes displacement pressure as a core feature of gentrification processes. Residential mobility decisions are therefore not random but emerge endogenously from heterogeneous household preferences and neighborhood conditions. Similar to Mauro, residential mobility is fully endogenous. However, the present model introduces environmental preferences as an additional driver of migration.

The simulation of the model evolves through **discrete time steps** representing annual urban development cycles. During each step (year), households evaluate residential utility, relocation decisions occur, neighborhood populations change, rents are recalculated and long-term rent adjustments are applied. Through repeated iterations, neighborhoods may transition from low-income areas toward increasingly affluent and environmentally attractive districts. The resulting population composition continuously changes due to migration flows between income groups. In this context, it is important to note that agents are unable to enter or exit the system or the city. It is evident that all agents are confined to a closed system, thereby restricting their mobility to individual neighborhoods.

A major extension beyond Mauro is the explicit representation of **urban green spaces** in the gentrification model. Hereby, it is asserted that the designated urban green spaces are modeled as environmental amenities, with multiple characteristics affecting neighborhood attractiveness. These green spaces are referred to in the following as "parks". Each park possesses four central park attributes: park proximity, park size, park quality and park function. These attributes are derived from the green gentrification and environmental valuation literature, particularly from Rigolon [36] and Bottero [9]. Therefore, *park proximity* is defined as the distance between a neighborhood and the park in question. The model assumes that the presence of attractive green infrastructure in proximity to residential areas is associated with an increase in neighborhood desirability and housing demand. Consequently, green amenities have been found to contribute indirectly to rent escalation and socio-economic displacement. The parameter *park size* is used to denote the area in square meters, whereas *park quality* is a measurement of the overall condition of the park as perceived by the citizen. The quality of such areas can be influenced by various factors, including the level of biodiversity, the effectiveness of conservation efforts, and the availability of recreational and sporting activities. This brings us to the final park attribute, the *park function*. The indicator is intended to highlight the predominant role of the UGS for households residing in close proximity. The environmental attractiveness of each neighborhood is operationalized through a **MCDA** framework inspired by Caprioli [10] where all park attributes are aggregated into a composite neighborhood green attractiveness score which directly influences: household utility, migration decisions, rent levels and long-term neighborhood development. Unlike Caprioli, where MCDA primarily serves as a spatial assessment tool, the present model embeds MCDA within a fully dynamic agent-based simulation framework. Environmental attractiveness therefore becomes an endogenous driver of urban transformation processes.

In addition to the environmental characteristics of urban green spaces, the model explicitly represents the economic costs associated with park provision and maintenance. Each park is assigned one-time **investment costs** and recurring annual **operational costs**, both of which are calculated as a function of park size and quality. Larger and higher-quality parks require proportionally greater financial investments and maintenance expenditures, reflecting empirical assumptions from urban green infrastructure management. While investment costs are recorded as an aggregate indicator of the overall financial effort required for green infrastructure provision, annual operational costs may directly influence the housing market. These recurring costs are spatially distributed across surrounding neighborhoods using a distance-decay function, whereby neighborhoods located closer to a park receive a larger share of the maintenance costs. If cost allocation is enabled, the distributed operational costs are converted into monthly per-household charges and incorporated into local base rents. Consequently, neighborhoods benefiting most from high-quality green amenities also bear a greater share of their maintenance costs, slightly increasing housing costs over time. This mechanism complements the hedonic capitalization of environmental amenities by introducing an additional cost-based pathway through which urban green infrastructure contributes to residential affordability and neighborhood transformation.

In conclusion, it can be said that compared to Mauro, the present implementation substantially extends the environmental dimension of urban transformation processes. While Mauro primarily focuses on socio-economic feedback dynamics, the present model explicitly incorporates environmental amenities as a central mechanism driving residential sorting and housing market transformation. At the same time, the model operationalizes the MCDA-based environmental attractiveness framework proposed by Caprioli within a fully dynamic agent-based simulation environment.

3.2 Model Parameter

The values of the model parameter integrated in the green gentrification model are based on two sources. Firstly, it is evident that some of the values in question are derived from the gentrification model proposed by Mauro et al. [31]. Secondly, the present study is founded upon the subsequent research of Rigolon and Nemeth (2020) [36]. It is also important to note that certain parameters are contingent upon calibration results. The **grid parameters**, which represents the features of the model city, are based on the grid-layout of Mauro. In order to ensure a scale that is consistent with moderately large urban areas, the simulation utilises a 7x7 (9x9) grid. In addition, the number of agents (households) in the model city is derived from Mauro's paper. Within the simulation, 1,000 agents inhabit the city grid.

The **income distribution** of all household income groups is derived from data sourced from the 2022 USA Social Security Administration report [38], which is also utilised in the Mauro approach. In accordance with the available data, 38% of the agents are classified as constituting the 'low-income group', with an annual income of less than 29,999\$. The predominant proportion of the population, constituting 57%, is classified within the 'medium-income bracket', with annual wages ranging from 30,000 to 174,999\$. Finally, it is evident that the uppermost quintile of the population is comprised of the 'high-income group', with an annual income that exceeds 175,000\$.

Moreover, the methodology employed in the **calculation of rent** is grounded in empirical data drawn from the United States. The United Census Bureau asserts that the median gross rent in 2022 amounted to 1,300\$ per month [41]. This can be represented in the model by a *base-rent mean* of 7.2 and a *base-rent standard deviation* of 0.5. The parameter *demand_price_elasticity* is set to 0.08 and determines how strongly local housing demand influences rent levels within the simulation. More specifically, the parameter controls the sensitivity of rents to neighborhood occupancy pressure. A moderate value was selected to ensure gradual and realistic rent increases in highly demanded neighborhoods without producing excessive price volatility. Lower values weakened the relationship between demand and rent escalation, while higher values caused unrealistically rapid housing cost increases. Consequently, the chosen parameter value enables the emergence of stable long-term gentrification dynamics driven by localized competition for residential space.

Three additional parameters strongly influence the behavioral dynamics of the model and in particular the **agents preferences**: the *green_attraction* coefficient, the *move_search_radius* and the *utility_threshold*. These parameters determine the relative importance of UGS within household decision-making, the spatial flexibility of residential mobility, and the overall sensitivity of agents to unfavorable neighborhood conditions. Firstly, the parameter *green_attraction* is set to a value of 50.0 and represents the overall influence of environmental attractiveness on household utility. More specifically, the parameter controls the extent to which the neighborhood green score contributes positively to residential utility calculations. A comparatively high value was deliberately selected in order to ensure that UGS constitute a sufficiently strong driver of residential relocation and neighborhood transformation within the simulation. Preliminary calibration experiments demonstrated that lower values considerably reduced the influence of parks on household decision-making and consequently weakened the emergence of green gentrification dynamics. Conversely, excessively high values resulted in unrealistic concentration effects around green amenities. The chosen value therefore represents a balance between environmental influence and socio-economic affordability constraints. In practical terms, higher values of *green_attraction* increase the probability that higher-income households relocate toward environmentally attractive neighborhoods, thereby intensifying rent escalation and socio-spatial displacement processes. Secondly, the parameter *move_search_radius* is set to 4 and determines the maximum spatial radius within which households search for alternative residential opportunities. This parameter operationalizes bounded rationality and spatial limitations in real-world housing markets, as households are assumed to possess only limited information about distant neighborhoods. A radius of 4 was selected in order to allow sufficient residential mobility across the urban grid while still preserving localized migration patterns and neighborhood path dependencies. Smaller search radii led to highly static neighborhood structures and reduced overall relocation activity, whereas excessively large radii produced unrealistic levels of mobility and weakened the spatial clustering mechanisms central to gentrification processes. Consequently, the selected value creates a realistic balance between local neighborhood attachment and urban mobility. Finally, the parameter *utility_threshold* is defined as 20 and determines the minimum acceptable utility level before a household decides to relocate. This parameter therefore controls the overall sensitivity of agents to unfavorable residential conditions. A threshold close to zero would result in excessive relocation behav-

ior and unstable neighborhood dynamics, while very low thresholds would produce unrealistically immobile populations. The selected value ensures that households tolerate moderate dissatisfaction before considering relocation, thereby reproducing more realistic migration behavior. At the same time, the threshold enables the emergence of gradual displacement processes, since rising rents and declining affordability progressively push vulnerable low-income households below the acceptable utility level. Consequently, the parameter plays a central role in generating long-term gentrification dynamics within the simulation environment.

Urban green spaces are represented in the model by parks, which are defined by four different **park parameters**, and can be placed in every cell of the grid. But before that, the proportion of cells that become park centroids, is established by a fixed *park coverage* of 12%. This finding is derived from the research conducted by Rigolon et al. [36], who analyzed the park coverage of all major cities in the US. As previously stated, the parks are categorized according to various attributes, including *park size*, *park position* (proximity), *park function* and *park quality*. The fundamental model offers three distinct park dimensions: small (1,000m²), medium (10,000m²) and large (100,000m²). The positioning of the parks on the grid is determined at random. Furthermore, a distinction is made between three distinct *park functions*. Consequently, the parks are classified as either sports parks, recreational parks or greenway parks. The latter category is comprised of parks with an active transportation component that were built mainly between 2008 and 2015. Recent literature has indicated that the impact of each category on the process of gentrification is distinct. Sports parks have been found to exert minimal influence, while recreational and greenway parks have been observed to play a more significant role in this process, with greenway parks in particular demonstrating the most pronounced impact [36]. Therefore, the relative importance of these three functions is as follows: 0.1 for sports parks, 0.4 for recreational parks and 0.5 for greenways. This approach is informed by the distinctions outlined in the works of Bottero [9] and Rigolon [36]. The final park parameter that is considered is the *park quality*. Consequently, a value between 0.7 and 0.3 is assigned to each park randomly. Recent research has indicated that the location of parks is a significant factor in the risk of gentrification. The findings suggest that parks situated in closer proximity to downtown areas are more likely to trigger gentrification processes than those located on the outskirts of cities. Although *park size* is not a significant predictor, *park function* is of greater importance [36]. Consequently, an MCDA is conducted to calculate the green gentrification factor for each UGS using different weights: The weights assigned to each criterion are as follows: 0.4 for proximity, 0.1 for size and 0.2 for quality and function.

Furthermore, another central parameter of the model is the **hedonic capitalization coefficient** *beta_ugs*, which is set to a value of 0.2. This parameter determines the extent to which environmental attractiveness generated by Urban Green Spaces is capitalized into local housing prices. More specifically, *beta_ugs* controls how strongly the neighborhood green score increases the current rent level of a grid cell within the housing market equation. The implementation is based on the assumption from hedonic pricing theory that households are willing to pay higher rents for access to attractive green amenities, such as high-quality parks and greenways. The selected value of 0.2 was determined through iterative calibration experiments and literature-oriented plausibility considerations. Lower values reduced the influence of

Urban Green Spaces on rent formation to a level where environmental amenities no longer generated meaningful gentrification pressure. In such cases, residential mobility was primarily driven by socio-economic factors alone, resembling the dynamics of Mauro’s original model. Conversely, substantially higher values caused unrealistically rapid rent escalation around parks and produced excessive displacement dynamics within only a few simulation steps. The chosen value therefore ensures that green amenities exert a visible but not dominant influence on housing market transformation.

Finally, the model further incorporates a set of **park cost parameters** to represent the economic expenditures associated with the provision and maintenance of urban green spaces. Investment costs are specified by the parameter *cost_invest_per_m2*, which defines a one-time construction cost of 60\$ per square meter, while annual maintenance expenditures are represented by *cost_operational_per_m2_per_year*, set to 1.8\$ per square meter per year. Both parameter values are hereby derived from the average empirical estimates reported by Panduro et al. [34]. To account for the fact that higher-quality parks generally require greater financial resources, both cost components are adjusted by park quality. The parameters *quality_invest_multiplier* (0.5) and *quality_operational_multiplier* (0.3) scale investment and operational costs proportionally according to the normalized park quality score. Consequently, parks with higher environmental quality are associated with higher construction and maintenance expenditures, reflecting the increased costs of establishing and preserving high-quality green infrastructure. These parameters provide the economic basis for the cost module of the model and enable the simulation to capture both the financial investment required for park development and the long-term maintenance burden associated with urban green spaces.

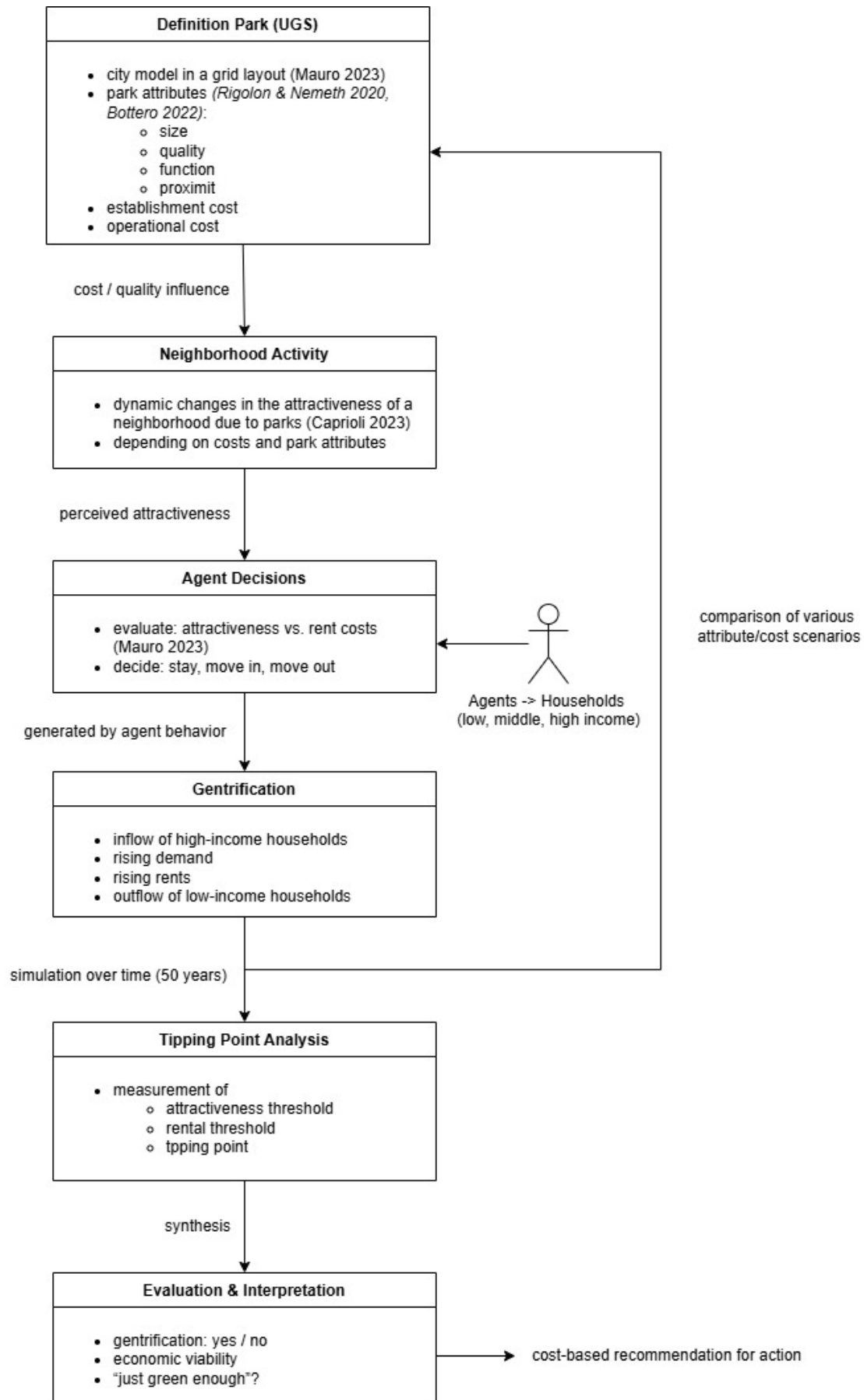


Figure 3.1: Model concept.

4. Implementation

This section presents the technical implementation and execution of the agent-based green gentrification model developed in this work. It describes how the conceptual model introduced in the preceding sections is translated into a computational simulation, including the underlying model architecture, agent behavior, spatial representation, and mathematical formulations governing residential mobility, rent dynamics, and the influence of UGS. In addition to the implementation of the individual model components, the section describes how the model is initialized and executed over discrete annual time steps, how household relocation and neighborhood conditions are updated throughout the simulation, and how relevant model outputs are collected for subsequent analysis. This provides the computational basis for reproducing the temporal and spatial dynamics of green gentrification and for evaluating the tipping behavior investigated in the experimental analysis.

4.1 Model Implementation

The model has been entirely implemented in *Python* (Version 3.14.5) using the *Mesa* agent-based modeling framework (Version 2.1.1)¹. It provides tools for the design, simulation and visualization of agent-based models, and offers a high degree of flexibility in the handling of complex data structures, modular model components and integration with advanced data analysis libraries [39]. *Mesa* is particularly well suited for the present study because it provides a comprehensive framework for implementing spatially explicit agent-based simulations while abstracting many low-level programming tasks. The framework offers standardized base classes for both models and agents, allowing complex simulation systems to be structured in a modular and object-oriented manner. Furthermore, *Mesa* includes flexible scheduling mechanisms that control the activation order of agents during each simulation step, thereby enabling the implementation of heterogeneous behavioral rules and decentralized decision-making processes. Its spatial modules provide different types of environments, including single grids, multi-grids, continuous spaces, and network structures. The present model utilizes a MultiGrid, which allows multiple household agents to occupy the same neighborhood cell simultaneously and therefore represents varying neighborhood densities in a natural way. In addition, *Mesa* provides efficient methods for agent placement, movement, neighborhood queries, and spatial interaction, substantially reducing the implementation effort required for residential mobility and local neighborhood effects. Another important feature is the integrated DataCollector, which automatically records user-defined model-

¹Access to the full code is provided via GitHub at: <https://github.com/Marco214/Master-Thesis>

and agent-level variables throughout the simulation. This facilitates the systematic evaluation of temporal developments, such as average rent levels, neighborhood attractiveness, and socio-economic composition. Mesa further supports the generation of interactive and static visualizations, enabling the spatial evolution of the model to be analyzed through maps and other graphical representations. Owing to its modular architecture, extensive spatial functionality, built-in data collection, and seamless integration with the scientific Python ecosystem, Mesa provides an appropriate and efficient platform for implementing the proposed green gentrification model. In addition to Mesa, several external Python libraries are used to support the implementation. *NumPy* [23] provides efficient numerical operations, random number generation, and array-based computations, while *Pandas* [44] is employed to import and process empirical datasets, particularly the household income distribution used for agent initialization. Furthermore, *NetworkX* [22] is utilized to support the representation and analysis of network structures where required, complementing the spatial modeling capabilities of the simulation framework.

Firstly, the 'GreenGentModel' class is derived from the 'mesa.model' class. The primary function of this class is to initialize and manage the agent-based simulation in its entirety. It is the model's function to create the spatial environment, generate household agents and UGS, manage model parameters, and oversee data collection throughout the simulation. Secondly, the 'household' class is defined as a subclass of the 'mesa.agent' class. The representation is of individual household agents within the simulation, with the agents characterized by their income group and income level. Moreover, it provides a utility-method that can be used to evaluate which neighborhood is the most suitable, and subsequently directs agents to that particular location.

The urban environment is represented as a two-dimensional 7x7-multi-grid in which each grid cell corresponds to a simplified neighborhood unit. Each cell stores information regarding its current residential population, the local rent level, and the environmental attractiveness generated by nearby urban green spaces. Multiple household agents may occupy the same grid cell simultaneously, allowing the emergence of spatial concentration and socio-economic clustering patterns. The model population consists of heterogeneous household agents divided into three income groups: low-income, middle-income, and high-income households. Each agent additionally possesses a continuous income value sampled from empirical income distribution data. This allows a more realistic representation of heterogeneous purchasing power and differentiated sensitivity to rising housing costs. Residential dynamics emerge from the interaction between households and neighborhood conditions. In each simulation step, corresponding to one year of urban development, agents evaluate the utility of their current residential location. If the utility falls below a predefined threshold, agents search for alternative neighborhoods within a limited spatial radius and relocate to the location with the highest expected utility. The full agent movement step can be seen in Algorithm 1.

Consequently, residential mobility is modeled as a bounded rational decision-making process under local spatial constraints. The utility function combines economic, environmental, and social dimensions of residential attractiveness. Formally, household utility U_i can be expressed as:

$$U_i = I_i + G_i + S_i - 40 * R_i$$

Algorithm 1 Agent Movement Step

```

1: currentPos  $\leftarrow$  agent position
2: currentCell  $\leftarrow$  cell at currentPos
3: currentUtility  $\leftarrow$  utility(currentCell)
4: if incomeGroup = high or currentUtility < utilityThreshold then
5:   bestCell  $\leftarrow$  currentCell
6:   bestUtility  $\leftarrow$  currentUtility
7:   candidates  $\leftarrow$  sampleCellsAround(currentPos, radius, 40)
8:   for all c  $\in$  candidates do
9:     u  $\leftarrow$  utility(c)
10:    if u > bestUtility then
11:      bestUtility  $\leftarrow$  u
12:      bestCell  $\leftarrow$  c
13:    end if
14:  end for
15:  if bestCell  $\neq$  currentCell then
16:    remove agent from currentCell
17:    add agent to bestCell
18:    move agent to bestCell.position
19:  end if
20: end if

```

where G_i denotes environmental attractiveness, S_i represents social neighborhood preference and R_i corresponds to rent burden. The disposable income component I_i is determined by the disposable income remaining after paying rent: $I_i = \frac{Income_i - Rent_i}{200}$. This term ensures that affordability remains a central determinant of residential decisions. Simultaneously, rent burden is modeled as the ratio of rent to disposable income. In addition, it is weighted differently for each income group:

$$R_i = \frac{Rent_i}{Income_i \cdot \sigma_g} \text{ with } \sigma_g = \begin{cases} 0.8, & \text{low-income} \\ 1.2, & \text{middle-income} \\ 2.5, & \text{high-income} \end{cases}$$

Higher rent burdens reduce overall utility and increase the probability of relocation, particularly for low-income households. Environmental attractiveness G_i enters the utility function through the neighborhood green score again weighted by income-group:

$$G_i = \omega_g \cdot \mu \cdot GreenScore_i \text{ with } \omega_g = \begin{cases} 0.3, & \text{low-income} \\ 1.0, & \text{middle-income} \\ 2.5, & \text{high-income} \end{cases}$$

where μ represents the general attractiveness coefficient of UGS for the hole model. This mechanism operationalizes the assumption that households prefer environmentally attractive neighborhoods with access to green amenities. The utility function additionally incorporates a social preference component S_I that captures socioeconomic neighborhood similarity as the fraction of neighbors of same group within Moore-neighborhood (the eight surrounding grid cells). Households derive additional utility from residing near agents belonging to the same income group, thereby reproducing endogenous socio-spatial clustering and segregation dynamics.

Housing prices evolve dynamically throughout the simulation and are determined endogenously by local occupancy pressure and environmental attractiveness. Current rent levels are calculated as:

$$Rent_i = BaseRent_i * (1 + \alpha D_i) * (1 + \beta G_i)$$

where D_i denotes local demand factor, G_i represents environmental attractiveness, α is the demand elasticity parameter, β captures the hedonic capitalization effect of UGS. Demand price elasticity is determined through neighborhood occupancy levels and operationalizes increasing competition for residential space. The second component represents hedonic pricing effects inspired by the environmental valuation literature. Neighborhoods with higher environmental attractiveness therefore experience systematically higher housing prices. In addition to short-term demand-driven rent variation, the model incorporates long-term rent drift. Base rents evolve over time according to: $BaseRent_{t+1} = BaseRent_t * (1 + 0.003D_i + 0.002G_i)$ This mechanism produces cumulative rent escalation in highly demanded and environmentally attractive neighborhoods, thereby reproducing long-term gentrification processes.

Urban Green Spaces constitute a central component of the model. Parks are implemented as explicit spatial entities distributed across the urban grid. Each park possesses four principal attributes derived from the green gentrification literature: park proximity, park size, park quality and park function. The influence of proximity p_i is modeled through an exponential distance decay function: $p_i = e^{-d/\lambda}$ where d denotes the euclidean distance between a neighborhood and a park, while λ controls the spatial decay rate. Consequently, the influence of green amenities decreases with increasing distance from the park. The park size s_i is normalized logarithmically in order to capture diminishing marginal utility effects: This ensures that larger parks generate slightly stronger attractiveness effects, while additional park area yields decreasing marginal increases in utility. The park quality is represented as a normalized value between 0.3 and 0.7 and reflects factors such as maintenance, usability, and aesthetic attractiveness. Higher-quality parks therefore generate stronger environmental utility and larger rent capitalization effects. The model further differentiates between different functional park types, namely recreational parks, sports parks, and greenways. Different park functions f_i receive different attractiveness weights based on their expected socio-economic impact. Greenways receive the highest weighting due to their strong association with urban redevelopment and green gentrification processes. Finally, the overall environmental attractiveness of each neighborhood is calculated using a MCDA framework inspired by Caprioli. The composite neighborhood green score is determined as:

$$GreenScore_i = w_p p_i + w_s s_i + w_q q_i + w_f f_i$$

where p_i represents park proximity, s_i denotes park size, q_i corresponds to park quality, f_i represents park function and w indicates the associated weighting coefficients. The resulting green score directly influences both household utility and housing prices. Consequently, UGS become endogenous drivers of migration, neighborhood transformation, and rent escalation. As demonstrated in Fig 4.1, the city-grid layout under consideration features multiple parks, along with their park attributes and associated green scores. Therefore, parks are represented by green circles. The diameter of each circle is proportionate to the size of the park in question, while

additional park attributes and costs are listed adjacent to the circles. The associated green score can be identified according to the darkness of the green of the cells, where darker cells have a higher green score than lighter cells.

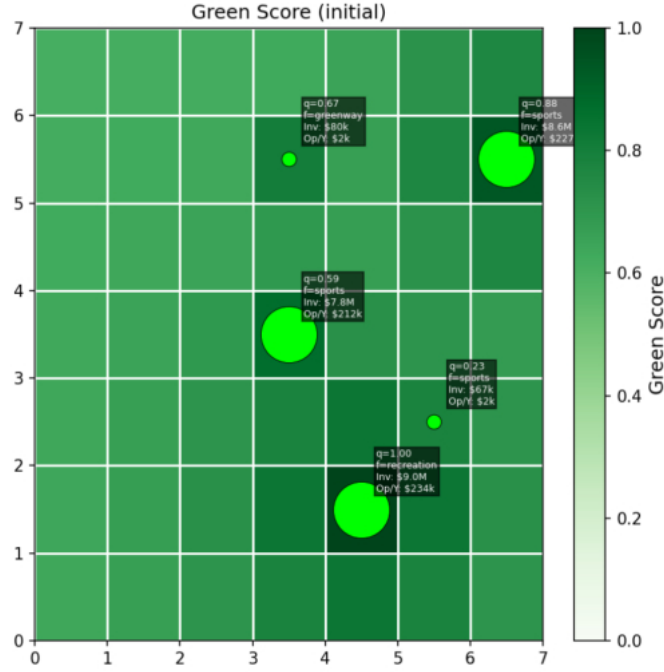


Figure 4.1: City example with parks and their green scores.

To complement the environmental valuation of Urban Green Spaces, the model additionally incorporates the financial costs associated with park provision and maintenance which can be manually activated by the boolean variable *enable_park_costs*. Each park is assigned both a one-time investment cost and recurring annual operational costs, which are calculated endogenously based on park size and park quality. Larger parks require proportionally greater financial resources, while higher-quality parks generate additional expenditures due to increased construction standards and maintenance requirements. The one-time investment cost of park p is calculated as

$$InvCost_p = s_p * c_{inv} * (1 + \alpha_{inv}q_p)$$

where s_p denotes the park size in square meters, c_{inv} represents the investment cost per square meter, q_p corresponds to the normalized park quality, and α_{inv} is the quality-dependent investment multiplier. Likewise, annual operational costs are determined as

$$OpCost_p = s_p * c_{op} * (1 + \alpha_{op}q_p)$$

where c_{op} denotes the annual maintenance cost per square meter and α_{op} specifies the quality-dependent operational multiplier. Consequently, higher-quality parks are associated with increased investment as well as higher long-term maintenance expenditures.

Within the simulation, investment costs are recorded as an aggregate indicator of the financial effort required to establish the urban green infrastructure. In contrast, annual operational costs directly affect the housing market. Similar to the modeling of environmental attractiveness, maintenance costs are distributed spatially

across neighborhoods using an exponential distance-decay function, whereby neighborhoods located closer to a park receive a larger proportion of its operational costs. The annual cost share allocated to neighborhood i is computed as

$$AnnualCost_i = \sum_p \frac{\exp(-d_{ip}/\lambda)}{\sum_j \exp(d_{jp}/\lambda)} \cdot OpCost_p$$

where d_{ip} denotes the Euclidean distance between neighborhood i and park p , and λ controls the spatial decay of park influence. The allocated annual costs are subsequently converted into monthly costs per household by dividing them by the number of households occupying the neighborhood and by twelve months. The resulting monthly surcharge is added directly to the neighborhood's base rent,

$$BaseRent_i^{t+1} = BaseRent_i^t + \frac{Cost_i}{12 * Occupancy_i}$$

before the demand- and amenity-based rent adjustment is applied. Consequently, the final rent paid by households reflects three interacting mechanisms: long-term rent appreciation, demand-induced rent increases, and the capitalization of environmental amenities, while also incorporating the recurring maintenance costs of nearby Urban Green Spaces. This additional cost module allows the model to represent not only the benefits of green infrastructure but also the financial burden associated with its long-term maintenance, thereby providing a more comprehensive representation of the economic processes underlying green gentrification.

4.2 Model Simulation

The execution of the agent-based simulation is controlled by the separate `simulation.py` module, which serves as the entry point for running experiments and evaluating model behavior over time. This script initializes an instance of the green gentrification model with a predefined set of model parameters, including the spatial grid size, number of household agents, park coverage, environmental weighting coefficients, and the hedonic pricing parameter. After model initialization, the initial spatial distribution of neighborhood characteristics and UGS is exported as a set of heatmaps, providing a baseline representation of the urban environment before any residential dynamics occur. The simulation then proceeds through a user-defined number of discrete time steps, with each step representing one year of urban development. During every iteration, the model executes one complete simulation cycle by calling the model's `step()` function, which sequentially updates household relocation decisions, neighborhood occupancy levels, housing prices, and all associated model variables. At regular intervals, defined by the `export_every` parameter, updated heatmaps are generated to visualize the spatial evolution of rents, incomes, population composition, and environmental attractiveness. Simultaneously, the built-in Mesa DataCollector records aggregated model statistics, including average rent levels and average neighborhood green scores.

To illustrate the spatial dynamics generated by the model, Figures 4.2 to 4.4 exemplarily depict the evolution of three central neighborhood attributes over the course of the simulation: average household income, monthly rent, and the local share of high-income households, each shown at year 0 (initial state), year 10, year 30 and

year 50. The underlying city configuration corresponds to the example introduced in Fig. 4.1, comprising five parks that differ in size, quality, and function: a large sports park of medium quality ($q=0.59$) in the center, a large sports park of high quality ($q=0.88$) in the upper right, a large recreational park of maximum quality ($q=1.00$) in the lower-central area, and two smaller parks — a high-quality greenway ($q=0.67$) in the upper area and a low-quality sports park ($q=0.23$) near the center-right — whose limited size and, in the case of the latter, low quality keep their environmental attractiveness comparatively low throughout the simulation.

The income heatmaps (Fig. 4.2) show a comparatively diffuse spatial income distribution at the initial state, with no discernible relationship to UGS locations. As the simulation progresses, however, a clear spatial association emerges between proximity to the two highest-quality large parks (the sports park in the upper right ($q=0.88$) and the recreational park in the lower-central area ($q=1.00$)) and above-average income levels. By year 10, distinctly yellow, high-income cells have already formed around these two locations, while the medium-quality large park in the center ($q=0.59$), despite being comparable in size, does not generate a comparably pronounced income cluster. This pattern intensifies by year 30 and remains largely stable through year 50, indicating that the system converges toward a new spatial equilibrium.

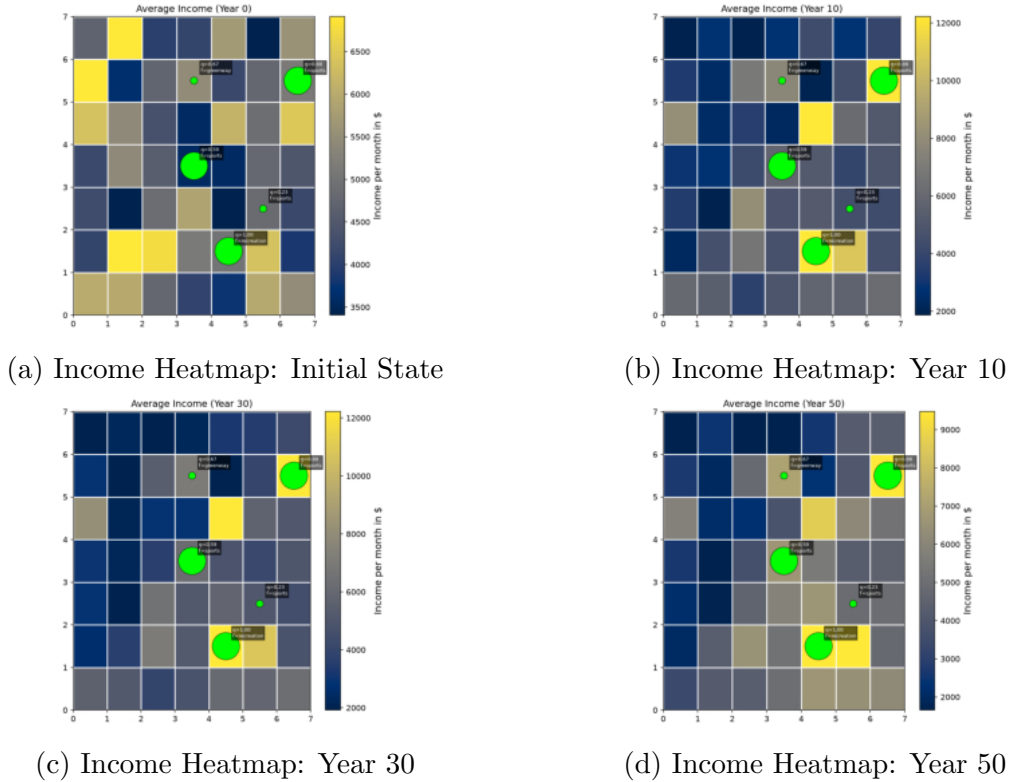


Figure 4.2: Income heatmaps from initial state to year 50 of the simulation.

A similar pattern emerges in the rent heatmaps (Fig. 4.3): high-rent clusters progressively form in the vicinity of the two high-quality parks, reflecting the hedonic capitalization mechanism implemented in the rent formation equation, whereby the neighborhood GreenScore, which is itself a function of park quality and park

function, weighted according to the MCDA scheme, directly increases local rents in proportion to the capitalization coefficient β_{ugs} . Notably, the rent response around the two attractive parks becomes visible earlier (already between year 0 and year 10) than the corresponding income redistribution, supporting the tipping-point framing developed for RQ1: rising, amenity-driven rents act as a comparatively fast, price-mediated displacement channel, whereas income-based sorting only stabilizes over the medium term.

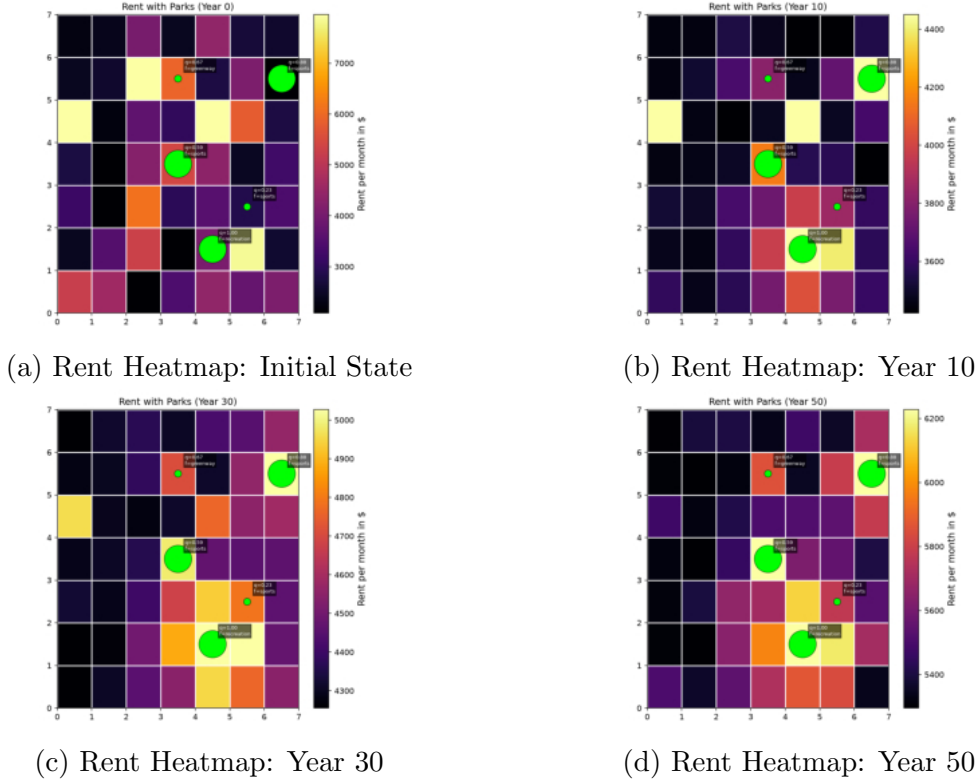


Figure 4.3: Rent heatmaps from initial state to year 50 of the simulation.

The high-income share heatmaps (Fig. 4.4) make the role of park quality and function particularly explicit. From year 10 onward, only the cells adjacent to the two highest-quality large parks (the $q=0.88$ sports park and the $q=1.00$ recreational park) are highlighted with a red border and exceed a high-income share of roughly 80%, while the medium-quality sports park in the center ($q=0.59$), although of identical size, never crosses this threshold despite hosting a visibly growing population cluster. This contrast illustrates, at the level of a single example run, precisely the result later confirmed systematically by the global sensitivity analysis in Section 5.2: park quality is the single most influential park attribute for the tipping probability, clearly outweighing park size, while park function also contributes a substantial total effect. This is consistent with the fact that both parks that do tip are functionally weighted more strongly in the gentrification-relevant literature (recreational and, to a lesser extent, sports use) than the smaller, low-quality park nearby ($q=0.23$), which never develops a visible high-income cluster at all. This concentration around the two high-quality parks persists through year 30 and year 50 and even intensifies slightly, while the remainder of the grid, including the vicinity of the medium- and low-quality parks, stays close to zero throughout.

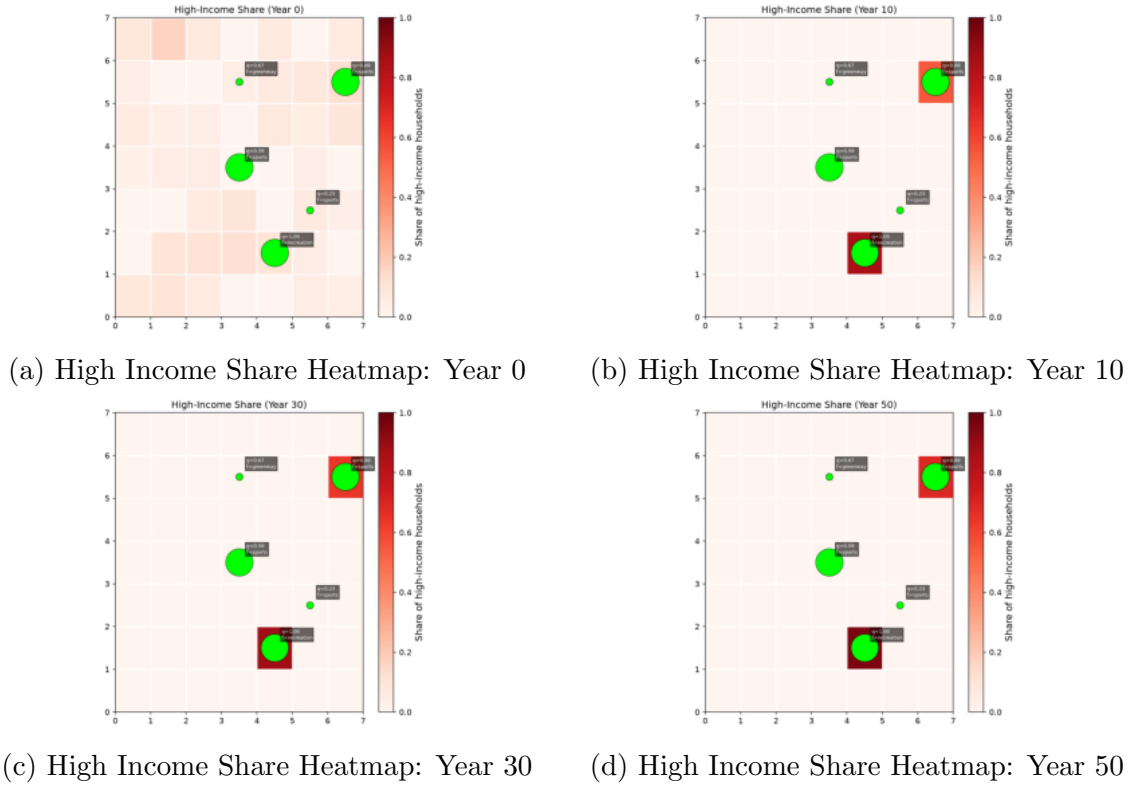


Figure 4.4: High income share heatmaps from initial state to year 50 of the simulation.

Taken together, the three heatmap series visually reproduce the central mechanism examined in this work: a temporally offset but causally linked sequence of rent appreciation, income-based redistribution, and ultimately stable spatial segregation around attractive green spaces. Crucially, this example run already foreshadows the central finding of the subsequent sensitivity and threshold analysis, which is that it is primarily park quality, and secondarily park function, rather than park size alone, that determines whether ecological enhancement tips over into a self-reinforcing green gentrification dynamic.

5. Experimental Evaluation

This chapter presents the experimental evaluation of the agent-based green gentrification model developed in the preceding chapters and provides the empirical basis for answering the two research questions. It first outlines the experimental design underlying both investigations, including the tipping-point criteria used to identify green gentrification, before presenting the corresponding results. A random seed analysis and a two-stage global sensitivity analysis first establish a statistically robust experimental setup and identify the park attributes most relevant for the tipping probability. Building on these preparatory steps, a full-factorial parameter sweep systematically examines how park quality, size, proximity and function jointly determine the timing and likelihood of green gentrification, addressing RQ1. Finally, a dedicated cost experiment investigates how establishment and maintenance costs shape these tipping dynamics, addressing RQ2 and the associated hypotheses H1 and H2. Together, these experiments provide a comprehensive assessment of both the socio-spatial and the economic dimensions of urban green space interventions within the proposed simulation framework.

5.1 Experimental Design

To systematically evaluate the role of urban green spaces in driving green gentrification, a structured simulation experiment was developed based on the proposed agent-based model. The experimental framework consists of two complementary investigations. First, a full-factorial parameter sweep systematically varies the key characteristics of urban parks to identify the conditions under which ecological improvements trigger self-reinforcing green gentrification dynamics. Second, a dedicated cost experiment evaluates how different establishment and operational cost scenarios influence these tipping points while keeping the underlying socio-economic mechanisms unchanged. Prior to both investigations, a random seed analysis is conducted to determine an appropriate number of Monte Carlo repetitions, and a global sensitivity analysis is performed to identify which park attributes warrant particularly detailed examination in the subsequent threshold analysis. Together, these preparatory and main experiments provide a comprehensive assessment of both the socio-spatial effects and the economic implications of urban green space interventions. A flowchart with an overview of the Experiment Design for RQ1 and RQ2 can be seen in Fig. 5.1.

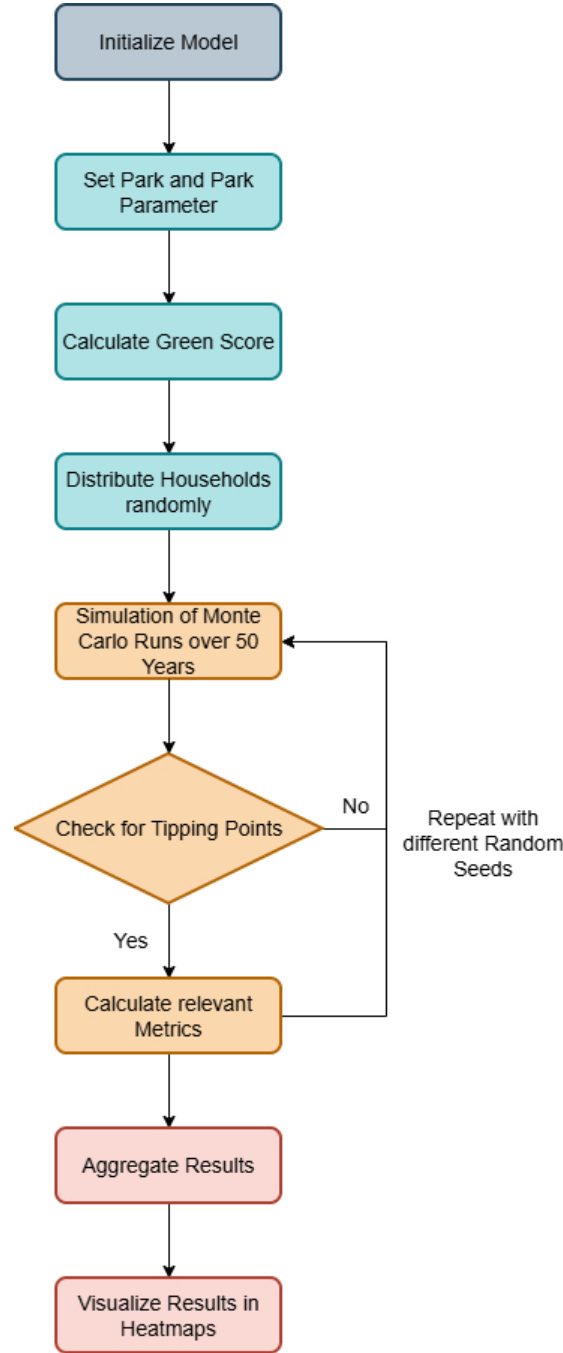


Figure 5.1: Flowchart of the experiment design for RQ1 and RQ2.

5.1.1 RQ1: Identifying Tipping Points

To address **RQ1**, which asks: **At what thresholds of park quality, size, or proximity does ecological enhancement transition into a self-reinforcing gentrification dynamic**, a parameter sweep is conducted in which the characteristics of a single urban green space are systematically varied. Therefore, the four key park attributes of *park size*, *park quality*, *park function* and *park proximity* are varied with different values, which can be seen in Tab. 5.1. For each combination of parameter values, the model is initialized with identical baseline socio-economic conditions while varying only the selected park characteristics of one park which is located in the center district of the city layout. Prior to each simulation run, the park is reinitialized using

the selected parameter combination, after which the environmental attractiveness of all neighborhoods is recalculated to consistently reflect the modified park characteristics. Furthermore, all household agents are randomly redistributed across the urban grid before the simulation starts. This prevents the results from being influenced by a particular initial spatial configuration and ensures that neighborhood transformation emerges endogenously from the interaction between heterogeneous households and the modified UGS.

Table 5.1: Parameter values used for the park attribute experiments.

Park Attribute	Possible Values
Park Size	1,000; 10,000; 50,000; 100,000m ²
Park Quality	0.2; 0.4; 0.6; 0.8
Park Proximity	0.5; 2.0; 4.0
Park Function	Recreation; Sports; Greenway

Before conducting the full-factorial parameter sweep itself, two preparatory experiments are carried out to determine an appropriate number of Monte Carlo repetitions and to identify which of the four park attributes warrant particularly detailed examination. Since the model contains stochastic elements, such as the initial spatial distribution and income realizations of household agents, the number of Monte Carlo repetitions per parameter combination directly affects the reliability of the resulting tipping-point statistics. To determine a suitable number of repetitions, a **random seed analysis** is conducted in which the complete 144-combination parameter grid is executed repeatedly for a range of candidate repetition counts, $N_{\text{runs}} \in \{5, 10, 20, 30, 50, 75, 100\}$. For each candidate value, the resulting distribution of the median tipping time is summarized using descriptive statistics (mean, standard deviation, quartiles, inter-quartile range, and the width of the 95% confidence interval), and the resulting runtime is recorded. The candidate value at which these statistics stabilize, i.e. beyond which further repetitions no longer yield a meaningful reduction in variability, is selected as the final number of Monte Carlo repetitions for both the RQ1 parameter sweep and the RQ2 cost experiment, thereby balancing statistical robustness against computational cost.

In addition, a two-stage **Global Sensitivity Analysis** is conducted prior to the full-factorial sweep in order to assess the relative influence of the four park attributes on the tipping probability, and to identify which parameters should receive particular attention in the subsequent threshold analysis. The Global Sensitivity Analysis (GSA) is implemented in Python using the SALib library [24] and operates on the same underlying simulation model as the full-factorial sweep. As SALib requires continuous input factors, the categorical parameter park function is encoded as a continuous variable on the interval $[0,3)$ which is subsequently mapped back to the categories *recreation*, *sports*, and *greenway* via a floor operation for each simulation run; the resulting sensitivity indices for this parameter are therefore interpreted as an approximation rather than a genuine continuous effect. For every sampled parameter combination, the model is evaluated using several stochastic replicates with independent random seeds, and the model output passed to SALib is the median tipping time across these replicates. If no tipping point is reached within a given replicate, the corresponding tipping time is conservatively right-censored at the simulation horizon (treated as a lower bound rather than an exact value), reflecting the

assumption that tipping would have occurred, if at all, only after the observation window.

In the first stage, a **Morris screening** is performed across all four park parameters using the Elementary Effects method, generating a set of trajectories over a discretized parameter grid. This yields $N(D + 1)$ model evaluations, where N denotes the number of trajectories and $D = 4$ the number of parameters. The resulting mean absolute elementary effect, μ^* , is used to rank the parameters by their overall influence on the tipping probability, while the standard deviation of the elementary effects, σ , provides an indication of nonlinearity and parameter interactions. In the second stage, a variance-based **Sobol analysis** is conducted for the top-ranked parameters identified by the Morris screening, using Saltelli sampling with a specified base sample size and including second-order interaction terms. The remaining, less influential parameters are held fixed at their reference values during this stage, substantially reducing the number of required model evaluations. The Sobol analysis decomposes the variance of the tipping probability into first-order effects (S_1), reflecting the isolated contribution of each parameter, and total-order effects (S_T), which additionally capture interaction effects with the remaining parameters. Together, the Morris screening and the Sobol analysis provide the methodological basis for prioritizing the park attributes examined in detail in the subsequent full-factorial parameter sweep.

The final experiments are executed as a full-factorial parameter grid in which the park attributes are systematically combined to a total of 144 possible combinations. Each parameter combination can be simulated multiple times using different random seeds to account for stochasticity (e.g. the income and spatial distribution of the households) within the agent-based model. To ensure reproducibility, every simulation is initialized with a deterministic random seed derived from a predefined base seed. Consequently, identical parameter combinations produce reproducible stochastic realizations, while different Monte Carlo repetitions explore alternative spatial and socio-economic initial conditions in a controlled manner. This Monte Carlo approach enables the estimation of the probability that green gentrification occurs under identical planning conditions.

The emergence of green gentrification is hereby identified through a threshold-based **tipping-point analysis**. Rather than relying on a single indicator, the model defines green gentrification as the simultaneous fulfillment of three conditions:

1. **Rent threshold:** The average neighborhood rent must exceed a predefined percentage of the initial average rent (110%), indicating a substantial increase in housing costs.
2. **Persistence criterion:** The increased rent level must remain above this threshold for a predefined number of consecutive simulation steps (3 years). This prevents short-term fluctuations from being misclassified as structural neighborhood change.
3. **Socio-economic change:** The local share of high-income households within the park neighborhood must increase by at least a specified threshold compared to the initial state, indicating a measurable socio-economic transformation of the residential population.

Only if all three criteria are fulfilled is a simulation classified as having reached a tipping point of green gentrification. For each parameter combination, the probability of reaching this tipping point (p_k), the time until the tipping point is reached (t_k), the corresponding rent level at the tipping point ($rent_k$), as well as the associated investment and annual operational costs of the park intervention are recorded. Following the completion of all Monte Carlo repetitions, the individual simulation runs are aggregated for each parameter combination. The probability of green gentrification is calculated as the proportion of simulation runs reaching the tipping point, while the median tipping-point time and median rent level are used as robust summary statistics to reduce the influence of stochastic outliers.

Since the parameter sweep comprises a large number of independent simulation runs, the experimental framework exploits parallel computing to reduce the overall execution time. Individual parameter combinations and random seed realizations are independent of one another and can therefore be executed simultaneously without affecting the simulation results. The experiments are distributed across multiple processor cores using Python’s multiprocessing capabilities, whereby each process executes a subset of the parameter combinations and stores the resulting model outputs independently. This embarrassingly parallel structure substantially decreases the computational effort required for the full-factorial experiment and enables a considerably larger number of Monte Carlo repetitions to be performed within a feasible runtime. For all experiments, the simulations are executed on a Windows 11 virtual machine with 256 GB of RAM and access to 64 CPU cores, hosted on a server equipped with an AMD EPYC 7713 processor. To improve computational robustness during these long-running experiments, intermediate results are periodically written to disk, allowing partially completed parameter sweeps to be recovered in the event of an interruption and facilitating continuous monitoring of simulation progress. Consequently, parallel execution increases the statistical robustness of the experimental results while maintaining reproducibility.

Finally, the aggregated simulation results are transformed into a set of heatmaps that visualize the relationship between the selected park attributes and the emergence of green gentrification. These visualizations allow threshold regions to be identified in which improvements in park characteristics are most likely to trigger self-reinforcing socio-spatial transformation processes and therefore provide the basis for answering the first research question.

5.1.2 RQ2: Cost Implications

To answer **RQ2, which examines how establishment and maintenance costs for urban parks influence the tipping points at which green investments exert a gentrifying effect**, the experimental framework follows the same general simulation procedure and tipping-point detection methodology as described for RQ1. Household behavior, residential mobility, rent formation, and the three tipping-point criteria remain unchanged, ensuring that differences in the simulation outcomes can be attributed exclusively to the economic characteristics of the park interventions rather than to modifications of the underlying agent-based model.

In contrast to the full-factorial parameter sweep of RQ1, the second experiment does not evaluate all 144 park configurations. Instead, four representative park scenarios are defined that combine different park sizes and quality levels with corresponding

investment and operational cost assumptions. Park proximity and park function are kept constant (proximity decay parameter = 2.0 and recreational park function) in order to isolate the influence of park costs on green gentrification dynamics. The cost scenarios are based on the average establishment and maintenance costs for urban green spaces reported by Panduro et al. [34]. Their study estimates average establishment costs of approximately 60\$ per m² and annual maintenance costs of 1.8\$ per m² per year. To investigate the sensitivity of the model to different financial conditions, these baseline values are complemented by lower-cost scenarios using 40\$ per m² for establishment costs and 1.0\$ per m² per year for annual maintenance costs and higher-cost scenarios using 80\$ per m² for establishment costs and 2.5\$ per m² per year for annual maintenance costs, resulting in four combinations of high and low investment and operational costs. Table 5.2 summarizes the evaluated scenarios.

Table 5.2: Cost scenarios evaluated for RQ2.

Park Size (m ²)	Park Quality	Inv. Cost (\$/m ²)	Op. Cost (\$/m ² /year)
100,000	0.8	80.0	2.5
100,000	0.4	80.0	1.0
10,000	0.8	40.0	2.5
10,000	0.4	40.0	1.0

For each scenario, the park-specific investment and operational cost parameters are assigned during model initialization. The establishment costs are applied once at the beginning of the simulation, whereas annual operational costs are accumulated throughout the simulation period. Both cost components are additionally adjusted according to the selected park quality, reflecting the assumption that higher-quality parks require larger initial investments as well as higher long-term maintenance expenditures. Consequently, every simulation run produces not only the socio-economic indicators used for tipping-point detection but also the total investment cost and annual operational cost associated with the implemented park. As in RQ1, each scenario is simulated repeatedly using different random seeds to account for stochastic variation in household income distributions and initial residential allocation. The resulting simulation outputs are aggregated by calculating the probability of reaching the green gentrification tipping point, the median time until the tipping point is reached, the rent level at the tipping point, and the associated investment and operational costs. This enables a direct comparison of the financial effort required for different green infrastructure interventions and their corresponding likelihood of inducing green gentrification.

The python code is using the same multiprocessing framework described for RQ1. Since all simulation runs are independent, the four cost scenarios and their Monte Carlo repetitions can be evaluated simultaneously across multiple processor cores. After completion, the individual simulation results are aggregated into summary tables and visualized as cost-response heatmaps, relating total investment costs and annual operational costs to both the probability of green gentrification and the median tipping-point time. This provides an integrated assessment of the trade-offs between economic expenditure and socio-spatial transformation.

5.2 Results

This section presents the results of the experimental evaluation and addresses the two research questions through a sequence of complementary analyses. The results first establish the robustness of the simulation setup and identify the parameters that most strongly influence tipping dynamics through a global sensitivity analysis. Building on these findings, the subsequent full factorial parameter sweep examines how different park attributes affect both the timing and probability of tipping, before extending the investigation to different cost scenarios. Together, these results provide a systematic assessment of the factors shaping green gentrification tipping and form the empirical basis for answering RQ1 and RQ2.

5.2.1 Random Seed Analysis

As the agent-based model contains stochastic elements, such as the initial spatial distribution and income characteristics of household agents, repeated simulations with different random seeds are required to obtain robust estimates for each parameter combination. The experimental design therefore uses Monte Carlo repetitions to account for stochastic variation and to estimate the probability and timing of green-gentrification tipping points under otherwise identical planning conditions. For the full-factorial parameter sweep, the influence of the number of repetitions on the stability of the simulation results was evaluated before determining the final number of runs. The experiment was therefore conducted with `N_runs` values of 5, 10, 20, 30, 50, 75, and 100. For each setting, the complete parameter sweep of all 144 park configurations was executed, and the resulting median tipping-point time (*median_kipp_time*) was evaluated using several descriptive statistics.

The results are summarized in Tab. 5.3. Overall, the results indicate that increasing the number of random-seed repetitions progressively reduces the variation of the estimated tipping-point time. The mean tipping time decreases from 31.20 simulation steps with five runs to 30.65 steps with 100 runs, while the standard deviation decreases from 3.81 to 2.78. A similar stabilization can be observed for the Interquartile Range (IQR), which decreases from 5.00 for five runs to 3.50 for 75 runs and remains unchanged at 3.50 for 100 runs. The 25th and 75th percentiles also show only minor changes between the higher repetition levels, indicating that the central distribution of the estimated tipping times has largely stabilized. The comparison between 75 and 100 runs is particularly relevant for determining the final experimental configuration. At 75 runs, the mean tipping time is 30.74 compared with 30.65 at 100 runs, while the median changes only from 30.75 to 30.50. Most importantly, the IQR remains unchanged at 3.50, indicating no further change in the spread of the central 50% of observations. The width of the 95% confidence interval decreases only slightly from 0.94 to 0.92, corresponding to a change of approximately 2%. This suggests that increasing the number of repetitions beyond 75 provides only a marginal improvement in the precision of the estimated tipping-point time. The remaining changes in the minimum and maximum values are similarly small, with the maximum decreasing from 38.00 to 37.00 while the minimum remains at 24.00.

Based on these results, `N_runs=100` is selected for the final parameter sweep. This number provides a sufficiently stable estimate of the tipping-point statistics while

Table 5.3: Random seed analysis of the tipping time (in years) for different numbers of runs.

N_runs	5	10	20	30	50	75	100
Mean	32.20	31.89	31.69	31.79	31.66	31.74	31.65
Median	32.00	31.50	31.50	31.50	32.00	31.75	31.50
Std. dev.	3.81	2.96	3.21	3.28	2.97	2.86	2.78
Min	23.00	23.00	25.50	22.50	25.50	25.00	25.00
Max	41.00	40.00	40.00	40.00	39.50	39.00	38.00
25% Q	30.00	29.50	29.13	30.00	29.50	30.00	30.00
75% Q	35.00	35.00	33.50	34.00	33.63	33.50	33.50
IQR	5.00	5.50	4.38	4.00	4.13	3.50	3.50
95% CIW	1.70	1.40	1.07	1.08	0.98	0.94	0.92
Runtime	00:11:52	00:24:22	00:52:31	01:17:04	02:11:10	03:16:53	04:22:59

avoiding an unnecessary increase in computational cost. Although additional repetitions could further reduce stochastic uncertainty, the computational runtime increases substantially with the number of simulations. The complete experiment requires approximately 3 hours and 17 minutes for 75 runs and 4 hours and 23 minutes for 100 runs. Thus, increasing the number of repetitions from 75 to 100 already results in an additional runtime of more than one hour, while producing only a minor change in the resulting statistical measures. Consequently, 100 runs are considered an appropriate compromise between statistical robustness and computational efficiency. This choice also provides a consistent basis for the subsequent full-factorial analysis. For each of the 144 parameter combinations, 100 independent Monte Carlo realizations are performed, resulting in 14,400 individual simulation runs for the RQ1 experiment. The resulting repetitions allow the probability of reaching a green-gentrification tipping point to be estimated as the proportion of runs in which the predefined tipping criteria are fulfilled, while the median tipping-point time provides a robust measure of the temporal development of the process.

5.2.2 Global Sensitivity Analysis

Before conducting the full factorial parameter sweep to answer RQ1, a Global Sensitivity Analysis (GSA) is performed to determine the relative importance of the four park attributes for the tipping probability. Since a detailed threshold analysis across all four parameters would be computationally expensive, the GSA serves to reduce the dimensionality of the parameter space and focus subsequent analysis on the most influential parameters. This is done in two stages: first, a Morris screening provides a computationally efficient first assessment of parameter influence and potential nonlinear effects; second, a Sobol analysis is conducted for the most influential parameters identified by the screening, allowing their effects to be decomposed into first-order and total-order (interaction) contributions. Together, these two steps provide the basis for the subsequent tipping-threshold analysis in the full factorial parameter sweep.

Morris Screening:

As the first stage of the global sensitivity analysis, a Morris screening was conducted to identify which of the four park-related parameters have the strongest influence on the probability of reaching the green-gentrification tipping point, p_{kip} . The purpose of this screening is to reduce the dimensionality of the subsequent sensitivity analysis and, more importantly, to identify the parameters that should be examined in greater detail when determining the tipping thresholds relevant to RQ1. Rather than quantifying the contribution of each parameter to output variance in detail, the Morris method provides an efficient first assessment of parameter influence and can additionally indicate nonlinear effects and interactions through the variation of the elementary effects. The screening considered the four park parameters *park size*, *park quality*, *park proximity*, and *park function* over their respective parameter ranges. A total of 100 Morris trajectories were generated using four grid levels. Since the Morris method requires $N(D+1)$ model evaluations for D parameters, this resulted in $100(4+1) = 500$ parameter combinations. Due to the stochastic nature of the agent-based model, each parameter combination was simulated five times using different random seeds. The resulting tipping probability (p_{kip}) was calculated as the proportion of replicates in which the predefined tipping criteria were reached. Consequently, the Morris analysis was based on 500 parameter combinations and 2,500 individual model runs in total.

The results are presented in Tab. 5.4 and a graphic representation is given in 5.2a. The parameters are ranked according to their mean absolute elementary effect, μ^* , which provides a measure of their overall influence on the tipping probability independently of the direction of the effect. Park quality exhibits the highest value with $\mu^* = 0.465$, clearly exceeding the values obtained for park function, park size, and park proximity, which all have $\mu^* = 0.270$. This indicates that, within the investigated parameter ranges, changes in park quality have the strongest overall influence on the probability that the model reaches a green-gentrification tipping point. Park function, park size, and park proximity show comparable levels of influence and are therefore considered secondary but still relevant factors. The standard deviation of the elementary effects, σ , provides additional information on whether the influence of a parameter varies across the parameter space. Figure 5.2b shows that park quality exhibits the highest value ($\sigma = 0.519$), together with the largest μ^* . This suggests that its effect on p_{kip} is not constant across different model states and parameter combinations, potentially indicating nonlinear behavior and/or dependencies on other model parameters. The remaining parameters show somewhat lower values of σ , ranging from 0.372 for park function to 0.382 for park size and park proximity. Thus, while their effects also vary across the parameter space, the variation is less pronounced than for park quality. The signed mean elementary effects, μ , should be interpreted with greater caution. The values range from (-0.135) for park quality to (0.120) for park size. Since μ^* is based on the absolute elementary effects, it is used as the primary criterion for the parameter ranking. In particular, the negative value of μ for park quality does not imply that park quality has no influence; rather, it indicates the prevailing direction of the estimated elementary effects under the parameterization used in the screening. The relatively small differences between the μ^* values of the three remaining parameters should also not be interpreted as evidence that their effects are exactly identical, as the corresponding confidence estimates indicate a degree of sampling uncertainty.

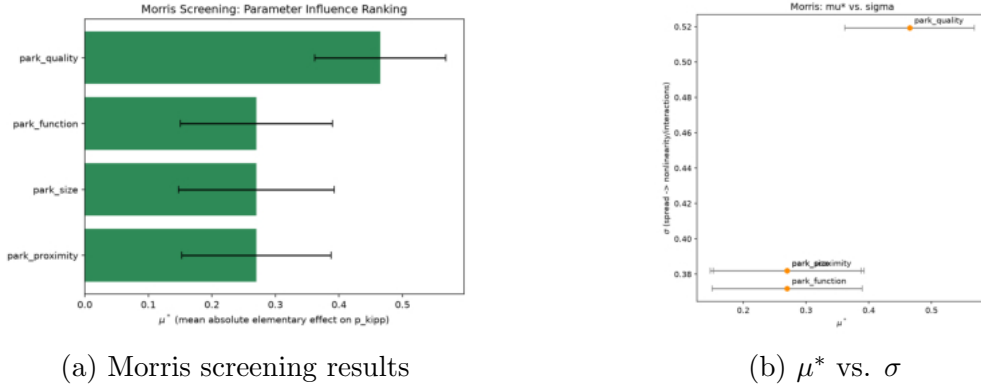


Figure 5.2: Results of the Morris Screening.

Table 5.4: Morris sensitivity analysis of the tipping probability.

Parameter	μ	μ^*	σ	μ_{conf}^*
Park Quality	-0.135	0.465	0.519	0.103
Park Function	-0.060	0.270	0.372	0.120
Park Size	0.120	0.270	0.382	0.122
Park Proximity	-0.120	0.270	0.382	0.118

Based on the Morris ranking, *park quality* and *park function* were selected as the two most influential parameters for the subsequent Sobol analysis. *Park size* and *park proximity* were therefore fixed at their reference values of 10,000m² and 2.0, respectively. This two-stage procedure substantially reduces the computational effort of the subsequent variance-based sensitivity analysis while retaining the parameters identified as most relevant by the screening. The results therefore provide a basis for focusing the subsequent analysis on the parameter dimensions that are most likely to affect the occurrence of a tipping point.

With respect to RQ1, the Morris screening provides an important first step towards identifying the conditions under which ecological enhancement transitions into a self-reinforcing gentrification dynamic. In particular, the results indicate that *park quality* should receive particular attention when determining tipping thresholds, while *park function* should also be considered as a potentially relevant conditioning factor. However, the Morris method does not itself identify the numerical threshold at which the transition occurs. Instead, it identifies the parameters for which a subsequent detailed threshold analysis is most informative. The subsequent Sobol analysis is therefore used to further investigate the contribution and potential interactions of the selected parameters, while the tipping-threshold analysis can subsequently examine how p_{kip} changes across their respective value ranges.

Sobol-Analysis:

As the second stage of the global sensitivity analysis, a variance-based Sobol analysis was conducted to quantify the contribution of the most influential parameters identified by the Morris screening to the variance of the tipping probability, p_{kip} . While the Morris method provides an efficient screening of parameter importance, the Sobol method allows the effects to be decomposed into first-order effects and

total effects, thereby providing a more detailed assessment of the individual contribution of each parameter and its contribution including interactions with other parameters. Based on the Morris ranking, *park quality* and *park function* were selected for the Sobol analysis, while *park size* and *park proximity* again were fixed at their reference values of 10,000m² and 2.0, respectively. For the two remaining parameters, a Sobol sampling design with a base sample size of $N = 512$ was generated. As second-order effects were included, the sampling procedure resulted in $N(2D + 2) = 512(2 \cdot 2 + 2) = 3,072$ parameter combinations. Each combination was evaluated using five independent stochastic model runs, resulting in a total of 15,360 individual model runs. For each parameter combination, the tipping probability p_{kip} was calculated as the proportion of stochastic replicates in which the predefined tipping criteria were reached. The resulting output was then used to calculate the first-order Sobol index (S_1), the total-order index (S_T), and the second-order interaction index (S_2). The results of the main and total effects are presented in Tab. 5.5 and a graphic representation is given in 5.3.

The results show very high total-order effects for both parameters. Park quality exhibits a total-order index of $S_T = 0.991$, while park function reaches $S_T = 0.901$. This indicates that both parameters account for a substantial share of the variation in the tipping probability when their overall effects, including potential interactions, are considered. Park quality therefore remains the most influential parameter identified in the sensitivity analysis, consistent with the results of the preceding Morris screening. The high total-order effect of park function additionally indicates that the type of park use can substantially affect whether the system reaches the gentrification tipping regime. In contrast, the estimated first-order effects are very small, with $S_1 = 0.029$ for park quality and $S_1 = 0.008$ for park function. Furthermore, both estimates are associated with relatively large confidence bounds ($S_1^{\text{conf}} = 0.118$ and 0.121, respectively). Consequently, the results do not provide strong evidence for a robust independent additive effect of either parameter. Rather, the difference between the first-order and total-order indices indicates that the influence of the parameters cannot be adequately characterized by their isolated main effects alone. However, the second-order interaction index between park quality and park function is close to zero ($S_2 = -0.027$) and has a relatively large confidence bound ($S_2^{\text{conf}} = 0.151$). The negative point estimate should therefore not be interpreted as a negative interaction; given the uncertainty of the estimate, the analysis provides no robust evidence for a specific pairwise interaction between the two parameters.

Table 5.5: Sobol main and total effects for the tipping probability.

Parameter	S_1	S_1^{conf}	S_T	S_T^{conf}
Park Quality	0.029	0.118	0.991	0.139
Park Function	0.008	0.121	0.901	0.127

Overall, the Sobol results complement the Morris screening by confirming the high overall relevance of park quality and park function for the tipping probability, while showing that their effects cannot be straightforwardly interpreted as independent additive contributions. This is particularly relevant for RQ1, as the occurrence of a self-reinforcing gentrification dynamic appears to depend strongly on the characteristics of the implemented green intervention. However, the Sobol analysis does not

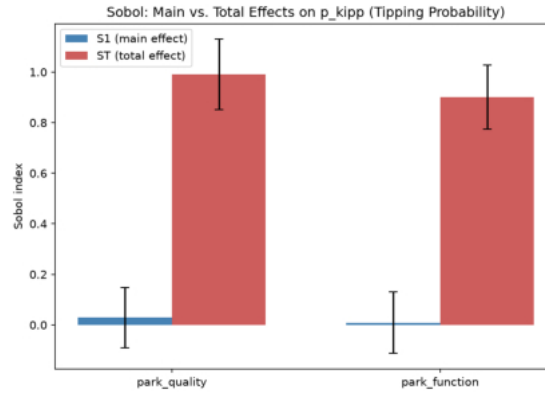


Figure 5.3: Graphical visualization of the Sobol main and total effects for the tipping probability.

identify the numerical values at which the transition occurs. Instead, it identifies the parameters for which a detailed threshold analysis is most relevant. The subsequent analysis can therefore focus on the response of p_{kipp} across the ranges of park quality and park function to identify parameter regions in which the probability of reaching the tipping regime increases substantially. In this way, the Sobol analysis provides the link between the global sensitivity assessment and the subsequent identification of the tipping thresholds in the full factorial parameter sweep required to answer RQ1.

5.2.3 Full Factorial Parameter Sweep

Before examining the individual influence of each park attribute, the aggregated statistics reported in Tab. 5.6 provide the relevant baseline against which the following marginal effects can be interpreted. Across all 144 parameter combinations, the median tipping time is 31.5 years, with a comparatively narrow inter-quartile range of 3.5 years, while the mean tipping probability is 0.167 with an IQR of only 0.043. This narrow overall spread, particularly for the tipping probability, which ranges from a minimum of 0.06 to a maximum of 0.26 across the full parameter space, already indicates that no single parameter combination produces extreme outlier behavior; rather, the observed variation results from the gradual, combined influence of all four park attributes. The marginal effects presented below decompose this aggregate variation into the contribution of each individual attribute.

To answer RQ1 in detail, the marginal (main) effects of each park attribute are computed by averaging the median tipping time and tipping probability across all combinations sharing the same value of the respective attribute, holding the other three attributes unconstrained. Tab. 5.7 summarizes these marginal effects. The results reveal a clear asymmetry between the two outcome measures relative to the baseline established in Tab. 5.6. With respect to the median tipping time, all four attributes exhibit a monotonic relationship, but with markedly different shapes. Park quality produces the steepest initial decline (-1.72 years between quality 0.2 and 0.4), after which the marginal reduction diminishes to -0.97 and -0.94 years for each subsequent 0.2-increment, indicating a pattern of diminishing returns rather than a sharp threshold. Park proximity shows the largest overall range (34.00 to 29.45 years) with an almost linear decline (-2.49 and -2.06 years per step), making it

Table 5.6: Overall baseline of the median tipping time (in years) and tipping probability across all 144 parameter combinations.

Statistic	Median Tipping Time	Tipping Probability
Mean	31.653	0.167
Median	31.500	0.170
Std. dev.	2.777	0.035
Minimum	25.000	0.060
Maximum	38.000	0.260
25% Quantile	30.000	0.147
75% Quantile	33.500	0.190
IQR	3.500	0.043
95% CI	[31.195, 32.110]	[0.161, 0.173]

the single most influential attribute for the timing of the tipping process among the four. Park size shows the same diminishing-returns pattern as quality, but far more pronounced (-0.76, -0.32, -0.24 years), which is consistent with the logarithmic normalization of park size implemented in the utility function. The simulation output thereby reproduces the underlying modeling assumption of decreasing marginal utility from additional park area. Park function confirms the weighting scheme derived from the MCDA: sports parks tip markedly later (33.07 years) than recreational (31.07) and greenway parks (30.81), mirroring the assumed low, medium and high socio-economic relevance of these functions in the literature [36].

Table 5.7: Marginal effects of park attributes on median tipping time (in years) and tipping probability.

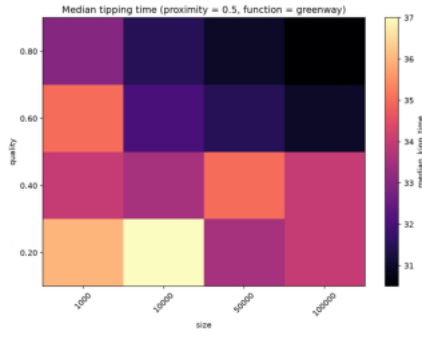
Parameter	Level	Median Tipping Time	Tipping Probability
Quality	0.2	33.67	0.170
	0.4	31.94	0.173
	0.6	30.97	0.158
	0.8	30.03	0.167
Size (m ²)	1,000	32.44	0.157
	10,000	31.68	0.172
	50,000	31.36	0.169
	100,000	31.12	0.170
Proximity	0.5	34.00	0.163
	2.0	31.51	0.168
	4.0	29.45	0.171
Function	Greenway	30.81	0.167
	Recreation	31.07	0.168
	Sports	33.07	0.167

This proximity effect and its interaction with park quality are directly visible when comparing the **median-tipping-time** heatmaps for greenway parks at the two extreme

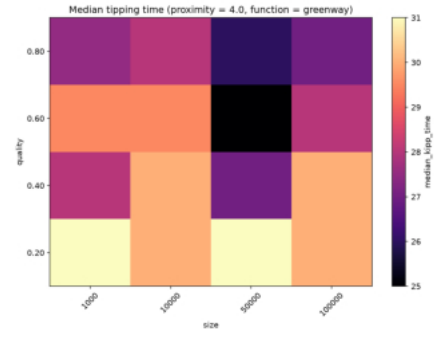
proximity settings (Fig. 5.4a and Fig. 5.4b). At $\lambda = 0.5$, tipping times range from approximately 31 to 37 years and the fastest-tipping cell is not consistently found at the highest quality level, reflecting the still-modest overall influence of environmental attractiveness when its spatial reach is narrow. At $\lambda = 4.0$, the entire color scale shifts downward to a range of roughly 25 to 31 years, and the fastest cells cluster more clearly in the upper-quality rows, particularly for larger park sizes. This side-by-side comparison illustrates both effects established in Tab. 5.7 simultaneously: the near-linear downward shift driven by proximity, and the increasingly visible, but still non-uniform, contribution of quality once the park’s amenity effect reaches a sufficiently large number of surrounding neighborhoods. The function-related lag identified above is equally visible in Fig. 5.4c, which shows the median tipping time for sports parks at the same low-proximity setting ($\lambda = 0.5$) used in Fig. 5.4a. Compared to the greenway panel, the entire color scale is shifted toward substantially higher values (approximately 33 to 38 years rather than 31 to 37), and no clear quality-driven gradient is discernible, even the highest-quality, largest-size cell does not reach the tipping speed observed for comparable greenway configurations. This directly corroborates the function ranking obtained from the marginal effects (33.07 years for sports versus 30.81 for greenway) and is consistent with the low MCDA weight assigned to sports parks (0.1) relative to greenways (0.5).

In contrast, the **tipping probability** shows no comparable monotonic pattern for any single attribute. Quality, for instance, does not decrease or increase steadily ($0.170 \rightarrow 0.173 \rightarrow 0.158 \rightarrow 0.167$), and the range across all marginal means (0.157 to -0.173) is narrow relative to the range observed across individual heatmap cells in Tab. 5.6 (0.06 to -0.26). This confirms, at the level of the full-factorial sweep, the central finding of the preceding Global Sensitivity Analysis: the first-order Sobol indices for quality and function were close to zero ($S_1 = 0.029$ and 0.008 , respectively), while the total-order indices were close to one ($S_T = 0.991$ and 0.901). The marginal effects reported here are the direct empirical counterpart of this result, no single attribute reliably predicts whether tipping occurs, only how quickly it occurs once triggered. This absence of a clear marginal pattern is directly visible in the tipping probability heatmap for sports parks at high proximity decay (Fig. 5.4d), which is among the most irregular of all combinations: the highest probability cell (quality = 0.8, size = 1,000, $p_{\text{kipp}} = 0.26$) and the lowest (quality = 0.4, size = 100,000, $p_{\text{kipp}} \approx 0.12$) occur at comparable or adjacent quality levels, without a discernible gradient across size or quality. To further substantiate this asymmetry, the 15% of parameter combinations with the highest tipping probability ($p_{\text{kipp}} \geq 0.20$, $n = 22$) were compared against the 15% with the lowest ($p_{\text{kipp}} \leq 0.13$, $n = 22$). Both proximity = 4.0 and function = sports are simultaneously overrepresented in the high-probability group (45.5% each, versus a 33.3% baseline share) and in the low-probability group (40.9–45.5%). Rather than indicating a contradiction, this pattern shows that these attribute levels are associated with the highest variance in tipping probability: whether a park with high proximity decay or a sports function tips with high or low probability depends on which quality and size values it is combined with, rather than on the attribute itself. Fig. 5.4d provides direct visual evidence of exactly this pattern, and thereby illustrates that interaction effects, not main effects, govern the probability dimension of RQ1.

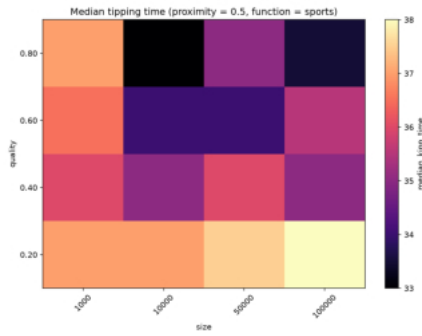
An additional illustrative example of such an interaction is given in Tab. 5.8, which reports the median tipping time for all four quality levels crossed with the three



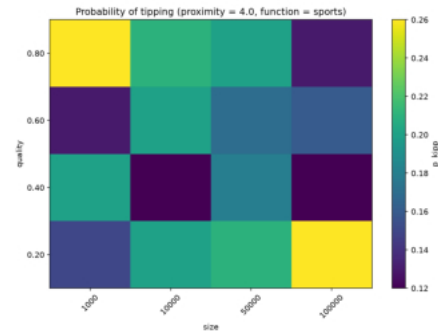
(a) Median tipping time for greenway parks at low proximity decay ($\lambda = 0.5$).



(b) Median tipping time for greenway parks at high proximity decay ($\lambda = 4.0$).



(c) Median tipping time for sports parks at low proximity decay ($\lambda = 0.5$). The systematically higher tipping times compared to Fig. 5.4a illustrate the function-related lag reported in Table 5.7.



(d) Tipping probability for sports parks at high proximity decay ($\lambda = 4.0$). The irregular, non-monotonic pattern illustrates the interaction-driven nature of the tipping probability discussed in the text.

Figure 5.4: Heatmaps illustrating tipping times and tipping probabilities across park functions and proximity-decay levels.

park functions, holding size (10,000 m²) and proximity ($\lambda = 2.0$) at their reference values. The quality effect is considerably stronger for greenways (33.0 $\rightarrow$ 28.0 years, a reduction of five years) than for sports parks (34.0 $\rightarrow$ 30.0 years) and is non-monotonic for recreational parks. This exemplifies why the isolated (first-order) contribution of quality to the variance of the tipping probability is small despite its high overall (total-order) importance: its effect is conditional on the co-occurring park function.

Taken together, the full-factorial results indicate that the timing of green gentrification tipping points is governed primarily by park proximity and, secondarily, by park quality and park function, each following a distinguishable but not sharply thresholded pattern. Quality and size show diminishing marginal returns, while proximity declines almost linearly across the investigated range. The occurrence of tipping, by contrast, cannot be attributed to any single attribute threshold, it emerges from the interaction between quality, function, size and proximity, consistent with the near-unity total-order Sobol indices reported. This distinction – a well-defined answer for *when* ecological enhancement is likely to tip, but only an interaction-based answer for *whether* it tips – is itself a substantive finding with respect to RQ1 and should be reflected in the discussion of planning implications in Chapter 6.

Table 5.8: Interaction between park quality and park function (median tipping time in years), at size = 10,000 m² and proximity $\lambda = 2.0$.

Quality	Greenway	Recreation	Sports
0.2	33.0	33.5	34.0
0.4	32.0	30.5	34.0
0.6	29.0	31.0	33.0
0.8	28.0	31.0	30.0

5.2.4 Cost Implications

Building on the experimental procedure outlined in Section 5.1.2, the second experiment evaluates how establishment and maintenance costs shape the tipping dynamics of green gentrification while holding park proximity and park function constant across the four representative cost scenarios summarized in Tab. 5.2. As in RQ1, each scenario was simulated repeatedly using independent random seeds, and the same threshold-based tipping-point criteria, comprising the rent threshold, the persistence criterion, and the socio-economic change condition, were applied to determine whether and when a simulation run reached the gentrification tipping point. The resulting median tipping-point time, tipping probability, and associated investment and operational costs for each of the four scenarios are reported in Tab. 5.9, while Tab. 5.10 summarizes the corresponding descriptive statistics across all cost combinations. Figure visualizes these results as cost-response heatmaps, relating total investment and operational costs to both the median tipping time and the tipping probability, and thereby provides the basis for the following interpretation of how the four cost profiles introduced in Fig. 2.2 relate to the model’s tipping behavior.

A first clear pattern emerges with respect to the median tipping time. As shown in Tab. 5.9 and visualized in Fig. 5.5a, the median tipping time decreases monotonically as total investment and operational costs increase, from 29 years in the Low/Low scenario (10,000m², $q=0.4$) to 25 years in the Low/High scenario (10,000m², $q=0.8$), 18 years in the High/Low scenario (100,000m², $q=0.4$), and finally only 10 years in the High/High scenario (100,000m², $q=0.8$). This ordering directly supports hypothesis H1, which assumes that higher establishment costs, typically associated with larger and higher-quality green infrastructure projects, lower the threshold of environmental attractiveness at which green gentrification emerges. The considerable spread between the fastest and slowest scenario, reflected in the comparatively large standard deviation of 8.347 and the wide 95% Confidence Interval (CI) of tipping time in Tab. 5.10, indicates that the size and quality of the cost-relevant park attributes exert a substantially stronger influence on the timing of the tipping process than on its probability.

The tipping probability itself follows a less linear pattern, as illustrated by the heatmap in Fig. 5.5b. Rather than increasing uniformly with cost, the tipping probability peaks in the Low/High scenario (10,000m², $q=0.8$) at 18%, followed closely by the High/High scenario at 17%, while the Low/Low and, most notably, the High/Low scenario yield the lowest probabilities of 15% and 11%, respectively. This pattern aligns well with hypothesis H2 and with the four cost archetypes in-

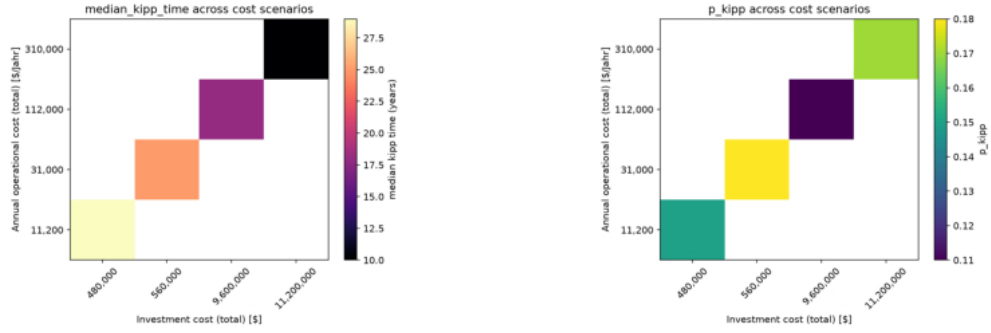
Table 5.9: Selected cost combinations (in \$) and corresponding tipping outcomes.

Size in m ²	Qual.	p_{kip}	Tipp. Time	Inv. Cost	Op. Cost	Cost Scenario
10,000	0.4	0.150	29 years	480,000	11,200	Low/Low
10,000	0.8	0.180	25 years	560,000	31,000	Low/High
100,000	0.4	0.110	18 years	9,600,000	112,000	High/Low
100,000	0.8	0.170	10 years	11,200,000	310,000	High/High

Table 5.10: Overall statistics of the median tipping time (in years) and tipping probability across all cost combinations after 100 runs.

Statistic	Median Tipping Time	Tipping Probability
Mean	20.500	0.152
Median	21.500	0.160
Std. dev.	8.347	0.031
Minimum	10.000	0.110
Maximum	29.000	0.180
25% Quantile	16.000	0.140
75% Quantile	26.000	0.173
IQR	10.000	0.033
95% CI	[7.219, 33.781]	[0.103, 0.202]

introduced in Fig. 2.2 The High/High scenario corresponds to representative, highly designed green spaces, which the literature associates with a high green gentrification risk driven by strong, immediate place-based amenity effects; the present results confirm this in terms of speed, as this scenario tips fastest of all four, even if its probability is marginally below the Low/High case. The High/Low scenario, by contrast, resembles large-scale structural projects, which are characterized by substantial upfront investment but comparatively moderate long-term upkeep; consistent with the literature’s framing of this profile as a source of predominantly short-term gentrification pressure, the model produces a comparatively fast but markedly less probable tipping process, suggesting that without sustained maintenance-driven quality preservation, the initial attractiveness effect of a large park does not translate into a persistently higher likelihood of tipping. The Low/High scenario mirrors socially oriented, intensively managed green spaces, for which the literature expects long-term, gradual effects; the simulation reproduces this pattern closely, as this scenario exhibits the second-slowest median tipping time while simultaneously achieving the highest overall tipping probability, indicating that sustained maintenance expenditure can compensate for a smaller establishment footprint by preserving park quality over time, in line with H2. Finally, the Low/Low scenario, corresponding to extensive, low-cost green spaces, produces the slowest tipping process of all four scenarios, broadly consistent with the literature’s characterization of this cost profile as posing a comparatively low green gentrification risk, even though its tipping probability is not the very lowest observed.



(a) Median tipping time through cost implications.

(b) Tipping probability through cost implications.

Figure 5.5: Heatmaps through cost implications.

Taken together, these findings suggest that establishment and maintenance costs affect green gentrification through two at least partially distinct mechanisms. Establishment costs, closely tied to park size and the associated immediacy of place-based amenity effects, primarily accelerate the speed at which tipping occurs once it is triggered. Maintenance costs, by contrast, appear to contribute more strongly to the sustained probability of tipping, since scenarios combining high operational expenditure with elevated park quality (Low/High and High/High) both outperform their low-maintenance counterparts in terms of tipping probability. This is further corroborated by the aggregate statistics in Tab. 5.10: the mean tipping probability across the four cost scenarios (0.152) lies close to, but slightly below, the mean tipping probability of 0.167 obtained across the full 144-combination parameter sweep reported in Tab. 5.6, while the variability in tipping time ($IQR = 10.000$) is considerably larger than the corresponding IQR of the full-factorial sweep (3.500). This indicates that, when proximity and function are held constant, cost-related park attributes exert comparatively limited influence on whether green gentrification eventually occurs, but a pronounced influence on how quickly it unfolds. In answering RQ2, these results therefore suggest that establishment costs primarily determine the tipping timeline, whereas maintenance costs, through their role in preserving park quality, are the more decisive factor for the long-term likelihood of a self-reinforcing green gentrification dynamic, this distinction closely mirrors the theoretical expectations formulated in hypotheses H1 and H2.

In summary, the experiments provide a coherent answer to both research questions. The analysis established 100 Monte Carlo repetitions as a robust simulation setting and identified park quality and function as the most influential parameters for tipping probability, with their effects largely driven by interactions. The full-factorial analysis further showed that tipping time responds monotonically to all four park attributes, with proximity having the strongest effect, while tipping probability depends on their combined interaction rather than on individual thresholds. The cost experiment confirmed this distinction: establishment costs primarily affected how quickly tipping occurred, whereas maintenance costs had a stronger influence on whether tipping occurred at all, supporting H1 and H2. Overall, the results distinguish between the timing and occurrence of green gentrification and provide the empirical basis for the policy discussion in Chapter 6.

6. Discussion and Outlook

This work developed an agent-based model of green gentrification that extends the dynamic gentrification framework of Mauro et al. [31] through the explicit integration of urban green spaces, inspired by the MCDA-based environmental valuation approach of Caprioli et al. [10], and, going beyond both, through an explicit representation of the establishment and maintenance costs associated with urban green infrastructure. Building on this framework, a structured experimental evaluation was conducted, comprising a random seed analysis, a two-stage global sensitivity analysis, a full-factorial parameter sweep, and a dedicated cost experiment, in order to systematically investigate the conditions under which ecological enhancement tips over into a self-reinforcing green gentrification dynamic.

With respect to RQ1 — at what thresholds of park quality, size, or proximity does ecological enhancement transition into a self-reinforcing gentrification dynamic — the results indicate that this question requires two distinct answers depending on which dimension of the tipping process is considered. Regarding the timing of green gentrification, all four park attributes exert a monotonic influence on the median tipping time, yet with clearly different characteristics: park proximity shows the strongest and most linear effect, while park quality and park size follow a pattern of diminishing marginal returns rather than a sharp threshold, and park function reproduces the literature-based ranking of greenway, recreational and sports parks. Regarding whether tipping occurs at all, however, no comparable threshold could be identified for any single attribute. Instead, the global sensitivity analysis and the full-factorial sweep consistently showed that the tipping probability is governed by interaction effects between park quality, function, size and proximity rather than by isolated main effects, as evidenced by the near-unity total-order Sobol indices and the absence of monotonic marginal trends in the tipping probability. RQ1 can therefore be answered as follows: a well-defined, quantifiable threshold pattern exists for when green gentrification is likely to occur, dominated by park proximity and, secondarily, by park quality, whereas whether it occurs at all cannot be reduced to a single attribute threshold but instead emerges from the joint configuration of all park characteristics.

Regarding RQ2 — how do establishment and maintenance costs for urban parks influence the tipping points at which green upgrades have a gentrifying effect — the cost experiment revealed a comparable asymmetry between timing and probability, but attributed to a different underlying mechanism. Establishment costs, closely tied to park size, primarily determined how quickly a tipping point was reached, with the median tipping time decreasing monotonically from 29 years in the Low/Low to only 10 years in the High/High cost scenario. Maintenance costs, by contrast, were more closely associated with the probability of tipping: scenarios combining

elevated operational expenditure with high park quality achieved the highest tipping probabilities, regardless of establishment cost level, while the scenario with high establishment but low maintenance costs produced a comparatively fast yet markedly less probable tipping process. These findings provide empirical support for both hypotheses formulated in Section 1.3: H1, that higher establishment costs lower the threshold of environmental attractiveness at which green gentrification emerges, is supported by the pronounced acceleration of tipping across the cost scenarios; H2, that higher maintenance costs shift green gentrification toward a more stable, long-term process, is supported by the elevated tipping probability observed whenever sustained maintenance expenditure accompanied high park quality.

Taken together, these findings directly address the problem statement and goal setting outlined in Chapter 1. Whereas the existing literature on green gentrification has predominantly relied on retrospective, empirical, or static valuation approaches that identify whether gentrification occurred without explaining the underlying feedback dynamics [9, 10, 36], the present agent-based framework reproduces green gentrification as an emergent process and thereby provides insight into how and under what conditions it unfolds over time. By explicitly incorporating establishment and maintenance costs into the simulation, a dimension that has remained largely absent from prior agent-based gentrification models, the framework further addresses the identified gap regarding the economic implications of urban green infrastructure investments, extending the analysis beyond ecological effectiveness toward long-term financial feasibility. Finally, the distinction identified between a comparatively predictable timing of tipping and a strongly interaction-dependent probability of tipping offers a first step toward operationalizing the normative JGE concept of Curran and Hamilton [16]: rather than relying on a single universal threshold, the results suggest that avoiding green gentrification requires attention to the specific combination of park attributes and cost structure, rather than the moderation of any single characteristic in isolation.

The results also provide several implications for urban planning and the design of green infrastructure interventions. Rather than defining universal threshold values at which green gentrification inevitably occurs, the findings suggest that planners should use park characteristics as indicators for a context-specific green-gentrification risk assessment. In particular, proximity, park size, quality, and function should be considered jointly when evaluating new green infrastructure projects. The results indicate that proximity has a comparatively strong and near-linear effect on the timing of tipping, while tipping probability is primarily driven by interactions between park characteristics rather than by a single parameter threshold. Consequently, combinations of highly attractive, large, and centrally located parks may warrant particular attention in neighborhoods that already experience increasing housing costs or socioeconomic change. The parameter combinations examined in this study can therefore serve as scenario-based reference points for identifying potentially high-risk configurations, rather than as universal policy thresholds. From a planning perspective, this suggests that the implementation of green infrastructure should be accompanied by measures that mitigate potential displacement effects. In areas with elevated gentrification risk, measures such as affordable housing provision, tenant protection, rent monitoring, and the preservation of existing affordable housing could be considered alongside park investments. Furthermore, the distinction between tipping time and tipping probability suggests that monitoring should not focus exclusively on whether

gentrification occurs, but also on how quickly socio-spatial changes develop following an intervention. Regular monitoring of rent levels, household composition, and residential mobility could therefore function as an early-warning mechanism. Finally, the cost results indicate that maintenance should be considered as part of the long-term social implications of green infrastructure rather than solely as an operational expense. Overall, the model is best understood as a decision-support framework that enables planners to compare alternative park configurations and identify situations in which ecological improvements may require additional social safeguards, rather than as a tool for establishing fixed thresholds that apply universally across different cities and neighborhoods.

Despite the insights provided by the model, several limitations should be considered when interpreting the results. First, the model is constrained by the availability and level of detail of the empirical data used for parameterization. The representation of urban green spaces is currently based on a limited set of characteristics, namely park size, quality, function, and proximity, while additional attributes such as accessibility, amenities, design, or maintenance intensity could further influence the relationship between green infrastructure and gentrification; future work could extend the MCDA-based attractiveness score accordingly. Similarly, household agents are primarily differentiated by income and therefore do not capture other potentially relevant characteristics such as age, gender, cultural background, household structure, or individual preferences, whose inclusion could reveal different relocation patterns and distributional effects. In addition, agents are currently restricted to the modeled urban system and cannot leave the simulation area; relaxing this closed-system assumption would allow displacement beyond city boundaries to be investigated. Second, the spatial and temporal scope of the model limits its ability to represent the complexity of real urban environments. The simulated city consists of a relatively small grid and a fixed number of household agents, while real cities are considerably larger and characterized by more complex spatial structures; scaling the simulation accordingly could reveal additional long-term or spatial dynamics beyond the 50-year horizon considered here. The representation of the housing market is also simplified, as rent formation and residential mobility are governed by a limited number of behavioral and economic mechanisms; factors such as housing supply, vacancy, property ownership, new construction, or regulatory interventions such as rent control are not explicitly represented, and their integration would allow policy measures to be tested directly within the simulation. A further limitation concerns the reliance on model assumptions and parameter choices. Although parameters are informed by existing literature and calibration experiments, several behavioral relationships and coefficients cannot be directly validated against observations from a specific real-world city, and different parameterizations, particularly regarding the influence of green amenities on residential utility and rent formation, could produce different tipping dynamics. The operational definition of a green-gentrification tipping point represents another important limitation, as tipping is identified through the simultaneous fulfillment of a rent threshold, a persistence criterion, and a socio-economic change criterion, and alternative thresholds or definitions could lead to different tipping probabilities and times. Future work should therefore apply the framework to an empirically documented case of green gentrification, to validate simulated tipping times and probabilities against observed data and to test the robustness of the tipping-point definition itself. Finally, the economic

dimension is subject to simplification, as establishment and maintenance costs are represented through generalized parameters linked primarily to park size and quality, whereas real costs vary substantially with local land prices, labor costs, and design requirements. The cost experiment therefore provides insight into the mechanisms through which costs affect tipping dynamics within the model rather than a direct estimate for specific real-world projects; extending this experiment toward empirically grounded, city-specific cost data and a full factorial cost sweep analogous to the RQ1 analysis would allow this trade-off to be examined with comparable rigor. Taken together, these limitations indicate that the model is best suited for exploring relationships, interactions, and potential tipping dynamics, while further empirical data, greater model complexity, and validation against real-world cases would be required to increase its predictive and generalizability potential.

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Titel der Arbeit Agent-based modelling of green gentrification around parks and other urban green spaces taking into account cost implications

Studienfach Master of Sc. Wirtschaftsinformatik und Künstliche Intelligenz

Prüferin/Prüfer Prof. Dr. Ingo J. Timm

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